Which foraging strategy? Reading microscopic motility rules from a population release

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A population of cells that piles up where the food is does not thereby reveal how it got there: many different microscopic responses to a nutrient produce the same accumulation. We ask which of them a release experiment—drop a blob of cells and film it—can actually tell apart. Nine elementary responses, from tumbling less while headed up-gradient to simply swimming slower in rich patches, fit inside one model of position and heading, which we coarse-grain to equations for cell density and mean heading. Four numbers a film readily yields—where the cloud goes, how wide it gets, how elongated, and which way it points—obey closed equations, and each response enters them at its own power of time. This sorts the nine into six distinguishable groups and sets how long each must be watched. Read once the cloud has stopped moving, the same numbers separate two coefficients that are routinely conflated: the cloud's width returns the chemotactic mobility, its elongation returns the drift of its centre of mass, and the two can differ in sign. It is the mobility, not the drift, that fixes how much nutrient the population eats and how densely it accumulates—and it does so on the food's diffusive time rather than the dish's, so foraging finishes long before the density settles. Tracking single cells over a fixed lag supplies the remaining combinations and separates navigating well from moving informatively: the two responses that never carry a cell up-gradient are the two its motion reveals the most about. We tabulate what each measurement returns, and verify every prediction against particle simulations and a conservative finite-volume solution with no fitted transport coefficient.

I. INTRODUCTION

Suppose you find a dish of bacteria heaped up on a patch of food. What have you learned? Less than you might think.

A cell has more than one knob to turn. More nutrient may make it swim faster, jiggle more, hold its heading longer, tumble less often, or bias the turns it does make—and any one of these can end with the population sitting where the food is. Some of them involve no sense of direction whatever: a cell that merely slows down in a rich patch will accumulate there, for the same reason that traffic piles up where the road is slow. The heap is real, but it is not by itself evidence of chemotaxis.

The converse is just as awkward. A response that plainly changes how individual cells move can leave the population's centre of mass exactly where it started.

Two questions follow. *Forward*: how does each way of modulating motility change what the population does? *Inverse*—the one an experimenter actually faces—which of them can be told apart, from what measurement, and after how long? Run-and-tumble motion and its continuum limits supply a natural language for both [1–3]; related work has asked how information constraints shape discrete navigation strategies [4].

We answer both by fixing the nutrient landscape and following each response through to the population it produces. Nine elementary rules turn out to live inside a single model of position and heading (Sec. II), which we coarse-grain to equations for the two fields an imaging experiment can reconstruct—the cell density and the mean heading (Sec. III). Integrating those equations over space closes a small system for the handful of numbers a release experiment actually reports (Sec. IV).

Everything after that is a question of *when you look*. Early, while the cloud is still growing, each rule enters at its own power of time, and those exponents sort the nine into six distinguishable groups and fix how long each must be filmed (Sec. V). Late, once the cloud has stopped moving, the nine collapse onto a few transport numbers—among them two that are routinely conflated: a mobility that says where the cells end up and how much they eat, and a drift that says where their centre of mass went. The settled cloud's width returns the first and its elongation the second, and for a speed rule the two have opposite signs (Secs. VI–VIII). Finally, tracking single cells over a fixed lag supplies the combinations the fields merge, and separates navigating well from moving informatively (Sec. IX). Section X tests every prediction against particle simulations and a finite-volume solution of the coarse-grained equations.

The outcome is a field guide (Table IV, in Appendix E): each row a measurement you can make, and the mechanisms it distinguishes. A reader who wants the recipe rather than the derivation can start there and work backwards; Table I names the nine rules. Appendices A–H

carry the derivations, the propagator and the numerics.

II. NINE WAYS TO RESPOND

Our setting is deliberately spare: N noninteracting cells of one species in a planar disk of radius $\mathscr{R}$, swimming through a smooth, static, dimensionless nutrient field η (Fig. 2). The landscape shapes how the cells move; the cells neither deplete nor replenish it. Lengths are measured in $\mathscr{R}$ and times in $\mathscr{R}^2/\mathscr{D}_T^{(0)}$, so that $D_T^{(0)} = 1$ and $\vec{r}$ lives in the unit disk $\mathbb{D}$ (Appendix A).

A cell is a point $\vec{r} \in \mathbb{D}$ with a heading $\hat{n}(\theta)$, and it can read three things about its surroundings. The first is the *local level* $\delta\eta$, the nutrient excess over a uniform reference,

$$\delta\eta(\vec{r}) = \eta(\vec{r}) - \eta^{(0)}, \tag{1}$$

The other two are the components of the gradient resolved in the cell's own body frame: the *alignment signal* $\zeta_\parallel$ and the *steering signal* $\zeta_\perp$,

$$\zeta_\parallel(\vec{r},\theta) = \hat{n}(\theta)\cdot\vec{\nabla}\eta(\vec{r}), \qquad \zeta_\perp(\vec{r},\theta) = \hat{n}_\perp(\theta)\cdot\vec{\nabla}\eta(\vec{r}). \tag{2}$$

These three differ in what they can tell a cell. Turning in place leaves $\delta\eta$ unchanged, so a cell reading only the level knows how good its present spot is but nothing about which way to go. The alignment signal $\zeta_\parallel$ is positive when the cell is already headed toward food and negative when it is headed away, so it grades the current heading. The steering signal $\zeta_\perp$ says which way to turn to improve.

Four coefficients set the motion: the *speed* v, the *translational diffusivity* D_T (how much the path jiggles), the *rotational diffusivity* D_R (how fast the heading wanders), and the *tumbling rate* λ (how often the cell reorients abruptly). To first order about their reference values,

$$\begin{aligned} X(\vec{r},\theta) = X^{(0)} + X^{(\eta)}\delta\eta(\vec{r}) + X^{(\zeta_\parallel)}\zeta_\parallel(\vec{r},\theta), \\ X \in \{v, D_T, D_R, \lambda\}, \end{aligned} \tag{3}$$

where the parenthesized superscripts label *sensitivities*—how strongly one signal tunes one coefficient—not powers. Four coefficients times two signals gives eight responses. The steering signal cannot enter this way: it changes sign under a reflection that a scalar coefficient cannot see. It therefore acts through a ninth mechanism, *continuous steering* with signed sensitivity χ,

$$u_\theta(\vec{r},\theta) = \chi\,\zeta_\perp(\vec{r},\theta), \tag{4}$$

so that $\chi > 0$ turns a cell up-gradient. Case 9 is thus an angular drift, not a coefficient response.

The cast. Each numbered *case* switches on one sensitivity and sets the other eight to zero (Fig. 1 and Table I); we prescribe no signs, so a case says only *which* knob the nutrient turns, not which way. Two families divide them. **Level rules** (Cases 1–4) read only how much food is here, and a cell obeying one of them cannot tell up-gradient from down. **Aimed rules** (Cases 5–8) read whether the cell is currently pointed toward food, and **steering** (Case 9) turns it there directly.

TABLE I. **The nine response rules.** Each rule lets the nutrient tune one motility parameter, or (Case 9) steer the cell directly. *Level* rules respond to how much food is present and carry no directional information; *aimed* rules respond to whether the cell is currently headed toward food. The short names in the second column are used throughout in place of bare case numbers. Signs are free: the sensitivity may be positive or negative.

Case	Name (sensitivity)	What the nutrient changes
Level rules — read the local concentration $\delta\eta$		
1	level–jitter $D_T^{(\eta)}$	random positional jiggle
2	level–wander $D_R^{(\eta)}$	how fast the heading drifts
3	level–speed $v^{(\eta)}$	swimming speed
4	level–tumble $\lambda^{(\eta)}$	how often the cell tumbles
Aimed rules — read the heading-projected gradient $\zeta_\parallel$		
5	aimed–jitter $D_T^{(\zeta_\parallel)}$	jiggle, when pointed at food
6	aimed–wander $D_R^{(\zeta_\parallel)}$	heading drift, when pointed at food
7	aimed–speed $v^{(\zeta_\parallel)}$	speed, when pointed at food
8	aimed–tumble $\lambda^{(\zeta_\parallel)}$	tumble rate (*E. coli*)
Steering — reads the transverse gradient $\zeta_\perp$		
9	steering χ	turns the body up-gradient

Two of these are familiar. Case 8, tumbling less often while heading up a gradient, is the classical run-and-tumble strategy of *Escherichia coli* [1]; Case 9, turning the body toward the gradient, is what crawling eukaryotes such as *Dictyostelium* and neutrophils do. The other seven are equally admissible responses that a cell could in principle use, and the question of this paper is which of the nine an experiment can tell apart. That is not visible yet—it needs the coarse-graining of Sec. III.

All nine live in one Itô process for the position and heading,

$$\mathrm{d}\vec{r}_t = v\,\hat{n}(\theta_t)\,\mathrm{d}t + \sqrt{2D_T}\,\mathrm{d}\vec{W}_t, \tag{5a}$$

$$\mathrm{d}\theta_t = \chi\zeta_\perp(\vec{r}_t,\theta_t)\,\mathrm{d}t + \sqrt{2D_R}\,\mathrm{d}W_t^\theta + \mathrm{d}\mathcal{J}_t^{\mathrm{tum}} \pmod{2\pi}, \tag{5b}$$

with $\vec{W}_t$, W_t^θ independent Wiener processes and $\mathcal{J}_t^{\mathrm{tum}}$ a pure-jump process that resets the heading at tumbles. The corresponding Fokker–Planck equation, its no-flux wall condition and the turning kernel are given in Appendix A. One detail there decides signs later: the diffusive current is $-\vec{\nabla}(D_T P)$, not $-D_T\vec{\nabla}P$—invisible while D_T is constant, decisive once the nutrient makes it vary.

III. FROM CELLS TO TWO FIELDS

Tracking every cell's heading is more than any experiment reports, and more than we need. Three angular averages of the phase-space density P carry almost everything: the **density** ρ (how many cells are here), the **polarity** $\vec{p}$ (which way they point, on average), and the *nematic tensor* $\mathbf{Q}$ (whether they share an axis without

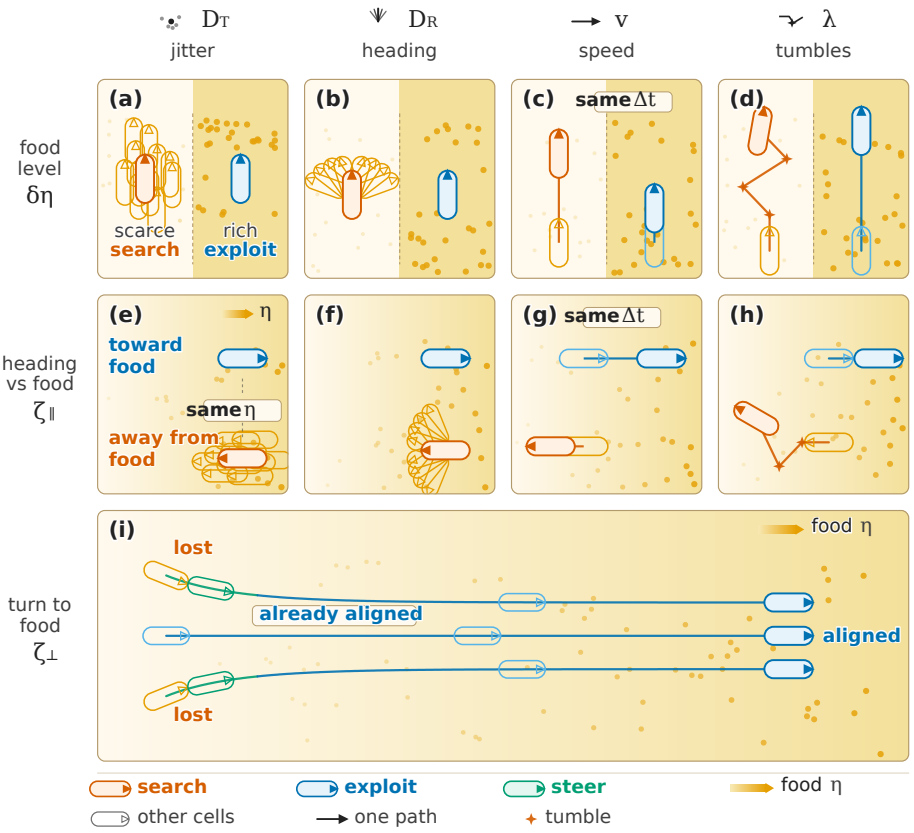


FIG. 1. **Nine ways a nutrient can change how a cell moves.** Gold shading is food; orange cells are in the searching state, blue in the exploiting state. *Rows:* what the cell reads. Cases 1–4 (a–d) read only the local level $\delta\eta$—rich patch versus poor—and so cannot tell which way to go. Cases 5–8 (e–h) read the alignment signal $\zeta_\parallel$, comparing a cell headed toward food with one headed away. Case 9 (i) reads the transverse signal $\zeta_\perp$ and turns. *Columns:* which of the four motility parameters is being changed—positional jitter D_T, heading wander D_R, speed v, or tumble rate λ. Case 8 is the classical *E. coli* strategy and Case 9 the eukaryotic one; see Table I for names. Trajectories are schematic and fix the sign convention only.

sharing a direction),

$$\rho = \int_0^{2\pi} P\,\mathrm{d}\theta, \qquad \vec{p} = \int_0^{2\pi} \hat{n}(\theta) P\,\mathrm{d}\theta,$$
$$Q_{ij} = \int_0^{2\pi} \left(n_i n_j - \tfrac{1}{2}\delta_{ij}\right) P\,\mathrm{d}\theta, \tag{6}$$

Density and polarity are the two fields an imaging experiment can actually reconstruct, so they are where we want the theory to live.

Integrating over θ gives an exact conservation law $\partial_t \rho = -\vec{\nabla}\cdot\vec{J}^{(\rho)}$, but not a closed one: density leans on polarity, which leans on $\mathbf{Q}$, and so on up an infinite chain. We cut the chain at $\mathbf{Q}=\mathbf{0}$—the *polar closure*—betting that alignment along an axis relaxes faster than transport can build it. Three explicit conditions control that bet (Appendix B), and Sec. X measures what it costs.

The rate at which a cell forgets its heading is what sets everything below. Rotational diffusion and tumbling both damp it, and they combine into a single *polarity relaxation rate* $\mathscr{D}_R^{(0)}$ (and its nematic counterpart $\mathscr{D}_{R,2}^{(0)}$),

$$\mathscr{D}_R^{(0)} = D_R^{(0)} + \lambda^{(0)}(1-\mathscr{P}_{\mathrm{turn}}) > 0, \tag{7}$$

$$\mathscr{D}_{R,2}^{(0)} = 4D_R^{(0)} + \lambda^{(0)}(1-\mathscr{P}_{2,\mathrm{turn}}) > 0, \tag{8}$$

so $1/\mathscr{D}_R^{(0)}$ is the time a cell stays pointed the same way—the natural clock of the problem. The nutrient shifts this rate through two combinations,

$$\mathscr{D}_R^{(\eta)} = D_R^{(\eta)} + (1-\mathscr{P}_{\mathrm{turn}})\lambda^{(\eta)},$$
$$\mathscr{D}_R^{(\zeta_\parallel)} = D_R^{(\zeta_\parallel)} + (1-\mathscr{P}_{\mathrm{turn}})\lambda^{(\zeta_\parallel)}. \tag{9}$$

These do different jobs. $\mathscr{D}_R^{(\eta)}$ is a change of tempo with no direction in it: cells forget their heading faster in richer patches, but nothing tells them where richer patches lie. $\mathscr{D}_R^{(\zeta_\parallel)}$ does carry direction—it converts the alignment signal into a genuine bias along $\vec{\nabla}\eta$. With the closure in hand, the angular moments close on

$$\partial_t \rho = -\vec{\nabla}\cdot\vec{J}^{(\rho)}, \tag{10a}$$
$$\partial_t p_i = -\partial_j J_{ij}^{(p)} + R_i^{(p)}, \tag{10b}$$

whose three constitutive pieces—the density flux, the polarity flux and the angular source—are written out in Appendix B.

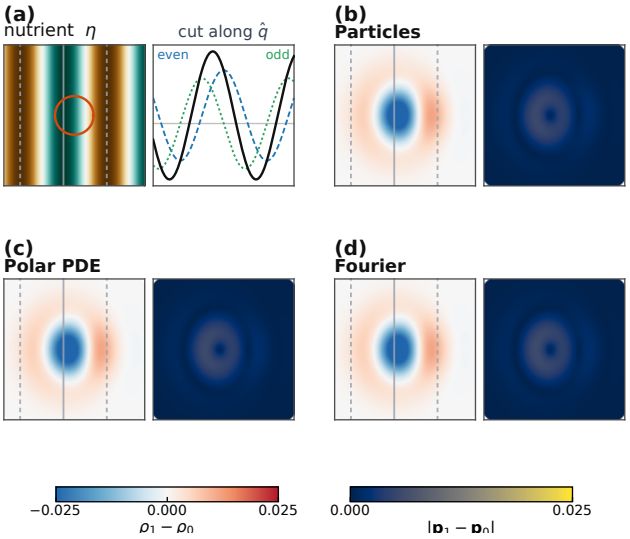


FIG. 2. **The experiment, and three independent calculations of it.** (a) The nutrient landscape: a single cosine, with the blob of cells released inside the orange circle. Solid lines mark food crests, dashed lines troughs, and the same marks reappear on every panel below, so the sign of a response can be read directly against them. The cut along $\hat{q}$ shows why the release sits off-crest: at this phase the landscape has equal odd and even parts about the release point, which excites the centroid and the spread at the same time. (b–d) The same problem solved three ways—by simulating individual cells, by integrating the coarse-grained equations, and by the closed-form Fourier solution—for level–jitter (Case 1) at $t = 5 \times 10^{-3}$. Each shows the density change $\rho_1 - \rho_0$ beside the polarity change $|\mathbf{p}_1 - \mathbf{p}_0|$. *What to look for:* the three columns should be indistinguishable, and they are, to $\sim 1\%$. Colour scales are shared with Fig. 7; level–jitter has the weakest polarity of the nine, which is why those panels look nearly blank.

Two patterns here drive everything below.

First, *the fluxes feel only the speed and jitter rules*—level–jitter, level–speed, aimed–jitter and aimed–speed (Cases 1, 3, 5, 7). The four turning rules and steering touch no flux at all; they enter through the angular source $\vec{R}^{(p)}$ alone. A turning rule can therefore move a population only by first tilting its polarity and then waiting for that polarity to carry cells—a detour that will cost it powers of time, and which is the origin of the whole hierarchy in Sec. V.

Second, aimed–speed (Case 7) is exceptional twice over: it pushes density directly, without polarizing anything first, and it alone manufactures nematic order out of the density.

IV. THE FOUR NUMBERS A FILM GIVES

The experiment we have in mind is a *release*: drop a compact blob of cells at one point of the landscape and film it. Few experiments resolve ρ and $\vec{p}$ everywhere; most report a handful of numbers per frame. This section shows that those numbers obey closed equations of their own, and that they suffice to identify the mechanism. We watch only while the cloud stays clear of the wall, $v_{\max} t + \sqrt{4 D_{T,\max} t} \ll 1 - r_0$.

With $\vec{x}$ the displacement from the release point, a film of the cloud yields four numbers per frame:

$$\vec{P} = \int \vec{p}\, \mathrm{d}^2x, \qquad \vec{X} = \frac{1}{N}\int \vec{x}\, \rho\, \mathrm{d}^2x,$$
$$S = \frac{1}{N}\int |\vec{x}|^2 \rho\, \mathrm{d}^2x, \quad T_{ij} = \frac{1}{N}\int \left(x_i x_j - \tfrac{1}{2}|\vec{x}|^2 \delta_{ij}\right)\rho\, \mathrm{d}^2x, \tag{11}$$

the *heading* $\vec{P}$ (which way the cloud points), the *centroid* $\vec{X}$ (where it has moved to), the *width* S, and the *shape* $\mathbf{T}$ (how elongated it is, and along which axis). Integrating Eqs. (10) over the plane turns two field equations into four ordinary differential equations for exactly these (Appendix E). The nutrient enters them only through six averages of the landscape weighted by the cloud, and through nothing else; no gradient expansion, adiabatic elimination or weak-landscape assumption has been made.

Two consequences are visible before any of it is solved, and both are diagnostic.

Parity. A landscape that is odd about the release point—food on one side—moves the population; an even one—a ridge or valley through the release point—only reshapes it. Which half of the landscape you release into decides which of the four numbers responds at all.

No flux can create alignment. The speed and jitter rules enter the equations only as fluxes, and a flux can push headings around without ever making them. So if a release develops a net heading, a turning rule or steering is at work, and the speed and jitter rules are excluded outright. That is the first entry in the readout below—and notice how much it buys for how little: one glance at whether the cloud points anywhere cuts nine candidates to five.

Reading it off the data

a net heading develops	a turning or steering rule; no flux rule can polarize ⇒ 2, 4, 6, 8, 9
no heading, but the centroid moves	a speed rule ⇒ 3, 7
neither heading nor centroid moves, ever	a jitter rule ⇒ 1, 5
only the width responds; the cloud stays round	a jitter rule again—a gradient feeds the trace alone
you released on a crest or in a valley	only width and shape can respond; move off-centre to see the centroid

V. TIMING: WHEN EACH RULE SHOWS ITSELF

Start at the beginning of the film. Different rules do not switch on at the same moment, and the reason is simple: a freshly released blob has density immediately but no polarity at all, so any rule that must work through polarity is delayed. Formally, the three landscape averages

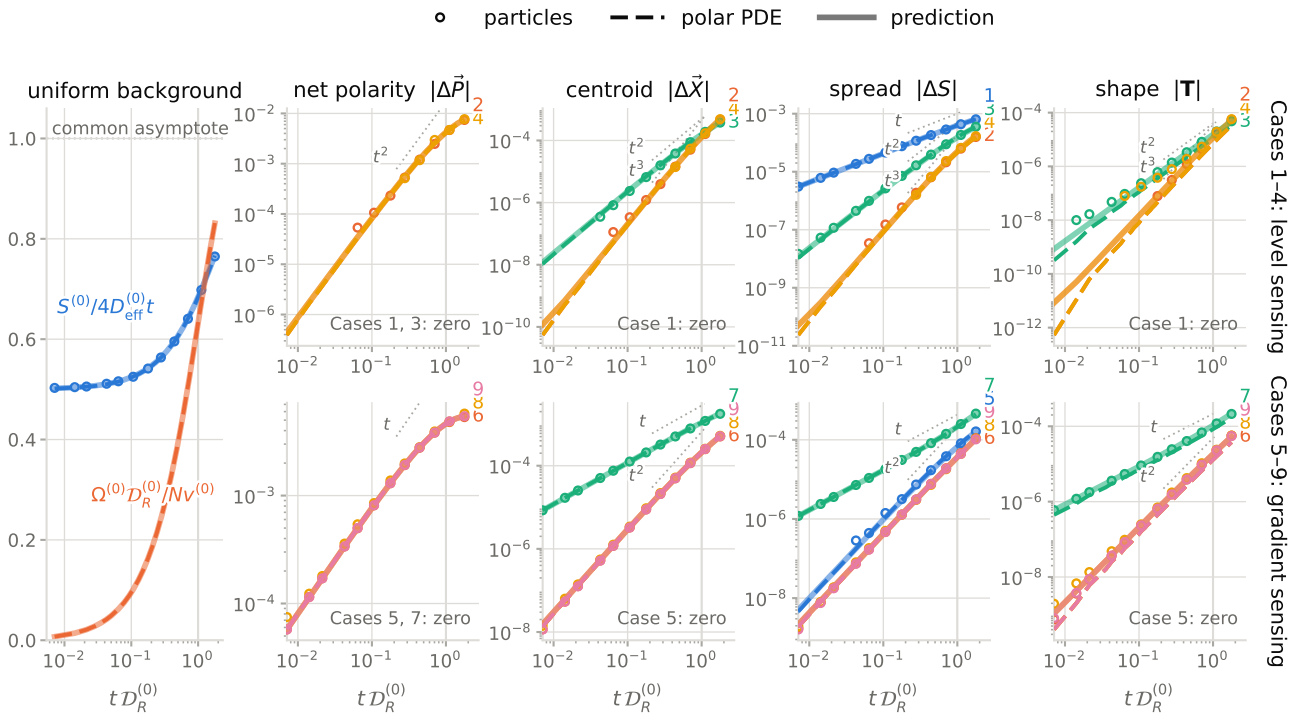


FIG. 3. **Each mechanism switches on at its own time, and the slopes identify it.** *Left:* the uniform-landscape baseline. The cloud spreads ballistically until cells forget their heading at $t\mathscr{D}_R^{(0)} = 1$, then diffusively; the outward alignment saturates at the same moment. One such curve calibrates $v^{(0)}$, $D_T^{(0)}$ and $\mathscr{D}_R^{(0)}$ before any nutrient is applied. *Right:* the response of each of the four observables, with the baseline subtracted, for the level rules (top row) and the aimed rules and steering (bottom row); curves are labelled by case number. *What to look for:* the slope on these log–log axes is the power of t, and the dotted guides are the predictions of Table II—so a curve running parallel to its guide confirms the mechanism's signature. Circles are simulated cells, dashed lines the coarse-grained solver, pale lines the closed-form theory. Markers are plotted only where a response exceeds twice its sampling error, so where a curve begins is the experimental resolution limit, not a physical onset.

weighted by density start finite, and the three weighted by polarity start at zero and grow like t.

One rule then generates the whole hierarchy: *the leading power of t is the power its driving overlap starts at, plus one for every time integration between that average and the observable.* Steering, for instance, reaches the heading through a density-weighted average, so $\vec{P} \sim t$; the heading then drives the centroid, so $\vec{X} \sim t^2$. Tracing each route through Eqs. (E3) gives Table II with no further calculation.

Result. *The exponents sort the nine rules into six distinguishable groups.* Two sets stay merged—level–wander with level–tumble, and aimed–wander with aimed–tumble and steering—and they are merged by the model itself, not by weak statistics: they enter only through the combinations $\mathscr{D}_R^{(\eta)}$ and $\chi - \mathscr{D}_R^{(\zeta_\parallel)}$. No ensemble size separates them.

This is a statement about experimental design, and a sharp one. Over a window $t \ll 1/\mathscr{D}_R^{(0)}$, the level–wander signal stands to the level–jitter signal as $\mathscr{D}_R^{(\eta)}(v^{(0)})^2t^2/D_T^{(\eta)}$—about one per cent for the parameters below. Film a release for a fifth of a heading-relaxation time and it will tell you almost nothing about how food shifts the turning statistics, no matter how many cells you count. Watching longer beats counting more.

Reading it off the data

signal grows as t	aimed–speed, or level–jitter in the width alone ⇒ 7, 1
signal grows as t^2	the rule needs one heading relaxation first ⇒ 3, 5, 6, 8, 9
signal appears only for $t \gtrsim 1/\mathscr{D}_R^{(0)}$, then as t^3	a level turning rule ⇒ 2, 4
nothing yet	you may simply be early: check $t\mathscr{D}_R^{(0)}$ before concluding the response is absent

VI. TWO COEFFICIENTS, NOT ONE

Wait long enough and the nine rules stop being nine. Once $t \gg 1/\mathscr{D}_R^{(0)}$, every cell has forgotten its original heading many times over; the polarity stops leading the density and starts merely following it, and what is left is textbook drift–diffusion, $\partial_t\rho = \vec{\nabla}\cdot[D_{\rm eff}\vec{\nabla}\rho - \varepsilon\kappa\rho\vec{\nabla}\eta_1]$. Only three numbers survive: the *effective diffusivity* $D_{\rm eff}^{(0)} = D_T^{(0)} + (v^{(0)})^2/2\mathscr{D}_R^{(0)}$ (how fast the cloud spreads when swimming is added to jiggling), its landscape-

TABLE II. **When each mechanism becomes visible.** Entries are the leading power of t with which a rule's signal grows in each observable, read off Eqs. (E3): a higher power means a later, weaker onset and a longer film. The four observables are the net polarity $\vec{P}$ (mean heading), centroid $\vec{X}$ (where the cloud went), spread S (how wide), and shape $\mathbf{T}$ (how elongated). A dash means *identically zero*, not merely small—a structural blind spot no ensemble size repairs. Reading the rows: aimed–speed is the fastest to appear, level–wander and level–tumble the slowest, and the two jitter rules are invisible in three of the four observables. For an idealized point release the Case 6–9 entries in S and $\mathbf{T}$, and Case 7's in $\mathbf{T}$, are delayed by one further power of t; all other entries are release-independent.

Case	Rule (sensitivity)	$\vec{P}$ heading	$\vec{X}$ centroid	S width	$\mathbf{T}$ shape
1	level–jitter $D_T^{(\eta)}$	—	—	t	—
2	level–wander $D_R^{(\eta)}$	t^2	t^3	t^3	t^3
3	level–speed $v^{(\eta)}$	—	t^2	t^2	t^2
4	level–tumble $\lambda^{(\eta)}$	t^2	t^3	t^3	t^3
5	aimed–jitter $D_T^{(\zeta_\parallel)}$	—	—	t^2	—
6	aimed–wander $D_R^{(\zeta_\parallel)}$	t	t^2	t^2	t^2
7	aimed–speed $v^{(\zeta_\parallel)}$	—	t	t	t
8	aimed–tumble $\lambda^{(\zeta_\parallel)}$	t	t^2	t^2	t^2
9	steering χ	t	t^2	t^2	t^2

dependent shift δD_{eff}, and the **chemotactic mobility** κ,

$$\kappa = \underbrace{\frac{v^{(0)}\big(\chi - \mathscr{D}_R^{(\zeta_\parallel)}\big) + \mathscr{D}_R^{(0)}\, v^{(\zeta_\parallel)}}{2\mathscr{D}_R^{(0)}}}_{\text{gradient responses}} - \underbrace{\left[D_T^{(\eta)} + \frac{v^{(0)}v^{(\eta)}}{2\mathscr{D}_R^{(0)}}\right]}_{\text{level responses}}. \tag{12}$$

Of these, κ is the one that decides where the cells end up: at steady state $\rho \propto \exp[\varepsilon\kappa\eta_1/D_{\text{eff}}^{(0)}]$, so $\kappa > 0$ means a heap on the food. Notice that the level rules contribute to it—the second bracket—and that this is no bookkeeping artefact. It is the traffic argument again, now with a coefficient attached: slower cells linger, lingering cells accumulate. In the limit of no jiggling it reduces to the familiar $\rho \propto 1/v$ [2, 3].

Now the trap, which is the central point of this paper. One expects the cloud's centre of mass to drift at $\kappa\vec{\nabla}\eta_1$—surely the coefficient that says where cells end up also says how fast they head there. It does not. Picture a blob straddling a region where the diffusivity varies: it spills faster to one side than the other, so its centre moves, and that displacement has nothing to do with where the blob will eventually settle. It adds on top. The centroid therefore obeys a *second, different* coefficient, the **pulse drift** $\mathscr{V}$,

$$\mathscr{V} = \frac{v^{(0)}\big(\chi - \mathscr{D}_R^{(\zeta_\parallel)}\big) + \mathscr{D}_R^{(0)}\, v^{(\zeta_\parallel)} + v^{(0)}v^{(\eta)}}{2\mathscr{D}_R^{(0)}} - \frac{\big(v^{(0)}\big)^2\mathscr{D}_R^{(\eta)}}{2\big[\mathscr{D}_R^{(0)}\big]^2}, \tag{13}$$

in which the level–jitter sensitivity $D_T^{(\eta)}$ has cancelled *exactly*. So level–jitter accumulates cells on food without ever moving the centre of mass, and—as Sec. VIII shows—it is also one of the rules that feeds the population best.

The two can point opposite ways, and the cleanest example is level–speed (Case 3): cells swim faster where food is richer. Then $\mathscr{V} > 0$ but $\kappa < 0$. Early on the cloud's centre moves *toward* the food, because that is the direction it spills fastest. Late on the population is heaped on the *poor* side, where cells crawl and dawdle. One rule, two opposite answers—because they answer two different questions. "Which way is it going?" and "where does it end up?" are not the same question, and this paper is largely about not confusing them.

Result. *A measured centroid velocity is not a chemotactic coefficient.* Report one and call it chemotaxis, and you can be wrong about the magnitude and about the sign.

Reading it off the data

the cloud drifts toward food	you have measured $\mathscr{V}$, and you do not yet know where the cells will end up
you want to know where they end up	measure the width, which gives κ
κ and $\mathscr{V}$ have opposite signs	a level speed rule ⇒ 3
$\kappa \neq 0$ but $\mathscr{V} = 0$	level–jitter ⇒ 1: accumulation with no drift at all

That said, three long-time numbers cannot carry nine rules, and waiting for steady state throws most of the information away: level–wander and level–tumble become indistinguishable, the three aimed-turning rules collapse onto one mobility, and aimed–jitter vanishes from all three numbers (Table III). Recovering the rest means reading the same film earlier (Sec. V) or one cell at a time (Sec. IX). What the late film *does* return, and how to read it off cleanly, is the next section.

VII. SATURATION: WHAT ONE RELEASE RETURNS

Two coefficients call for two measurements, and one film supplies both—but only once it has run long enough, and only if the right two moments are read off it. One practical warning first. A release in a periodic landscape does *not* drift steadily: once the cloud has spread across a wavelength the gradient it feels averages to nothing and all motion stops, after $\tau_\ell = 1/(D_{\text{eff}}^{(0)}q^2)$. *It reports a*

total displacement, not a velocity—fitting a slope gives an answer that depends on when filming stopped. What it does report is

$$\vec{X}(\infty) = \frac{\varepsilon\,\mathcal{V}}{D_{\text{eff}}^{(0)} q}\,\hat{q} \quad \text{(odd mode)}, \qquad \Delta S(\infty) = -\frac{4\varepsilon\,\kappa}{D_{\text{eff}}^{(0)} q^2} \quad \text{(even mode)}, \tag{14}$$

with $\Delta S = S - S^{(0)}$ the excess spread over a uniform landscape.

Result. *The centroid saturates on the drift; the width saturates on the mobility.* The two coefficients are separately readable off one release. The second line is worth pausing on: a population that accumulates on food ends up *narrower* than one in a uniform landscape—concentrating on the crests is literally a reduction in width, so measuring the width measures κ.

Better still, one parity suffices. The width uses only the trace of the second moment; its traceless part—the shape $\mathbf{T}$ of Eq. (11)—comes from the same images at no extra cost, and on an even mode it saturates on *the drift*,

$$T_{\parallel}(\infty) = -\frac{2\varepsilon\,\mathcal{V}}{D_{\text{eff}}^{(0)} q^2}, \tag{15}$$

with $T_{\parallel}$ measured along $\hat{q}$.

Result. *One landscape mode returns both coefficients at once, from the two parts of one tensor*: how wide the cloud got gives κ, how elongated it got gives $\mathcal{V}$. Where the two disagree, the population is being pushed by a position-dependent diffusivity rather than steered, and the size of the disagreement says how much.

Two properties make the shape the better of the two to measure. No jitter rule appears in it—a diffusive flux is a gradient, and a gradient feeds only the trace—so level–jitter and aimed–jitter cannot elongate a cloud at all. And a uniform landscape leaves the cloud exactly round, so unlike the width the shape carries no background to subtract: the whole measured signal is response. Appendix E adds the finite-width and finite-wavenumber corrections, and the heading plateau, which supplies a third independent combination.

Reading it off the data

final width, even mode	*returns* the mobility $\kappa = -D_{\text{eff}}^{(0)} q^2\,\Delta S(\infty)/4\varepsilon$
final elongation, even mode	*returns* the drift $\mathcal{V}$, with no background to subtract
final centroid, odd mode	*returns* $\mathcal{V}$ again—a consistency check
cloud stopped moving	you have waited long enough; stop filming at a few τ_ℓ
elongation stays zero	a jitter rule $\Rightarrow$ 1, 5

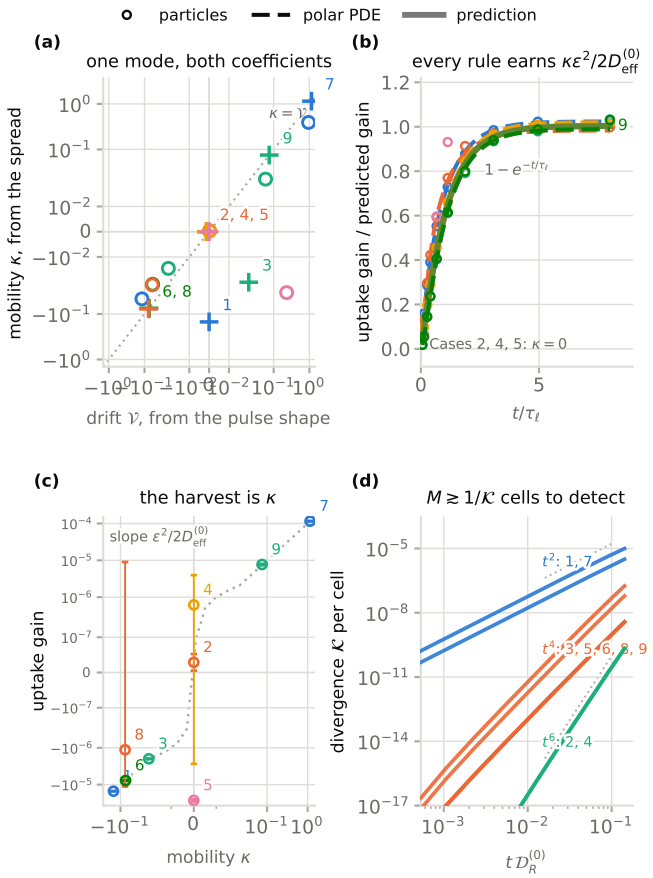


FIG. 4. **What one release measures, and what the population earns by it.** (a) Drift against mobility. Reading the width and the elongation of the settled cloud returns $\mathcal{V}$ and κ separately; circles are recovered values, crosses the values put in, and the dotted diagonal is where the two coefficients would agree. *The point of the panel is the points that lie off that diagonal*: level–speed (3) sits in the opposite quadrant—it drifts toward food while accumulating away from it—and level–jitter (1) sits on the vertical axis, accumulating strongly with no drift whatever. (b) Uptake over time, each rule divided by its own predicted gain; all collapse onto $1 - e^{-t/\tau_\ell}$, so a single landscape clock governs feeding. (c) The harvest law itself: nutrient actually collected against the mobility κ put in, falling on one line. Level–wander (2) and level–tumble (4) sit at the origin and earn nothing. Aimed–jitter (5) is the exception that proves the rule—it earns a real gain with $\kappa = 0$, because it feeds at one order beyond this expansion. (d) How distinguishable each rule is from a non-responding forager; $1/\mathcal{K}$ is the number of cells a detection needs, and the grouping is by the exponent each rule inherits.

VIII. FORAGING: WHAT THE POPULATION EARNS

Transport moments say where a population goes. A foraging experiment asks a different question—what does it actually eat?—and the answer turns out to be governed by κ, not by the drift $\mathcal{V}$. Define the **uptake** $\mathcal{U}$ as the

nutrient concentration the average cell finds itself in,

$$\mathcal{U}(t) = \frac{1}{N}\int \rho\,\eta\,\mathrm{d}^2x. \tag{16}$$

The wall is impermeable, so $\mathcal{U}$ changes only through the flux running along the landscape. Once alignment is quasi-static and the landscape is coarse compared with a run,

$$\dot{\mathcal{U}} = \underbrace{D_{\text{eff}}^{(0)}\langle\nabla^2\eta\rangle}_{\text{passive}} + \underbrace{\kappa\langle|\vec{\nabla}\eta|^2\rangle}_{\text{harvest}} + \mathcal{O}(\varepsilon^2). \tag{17}$$

The two terms differ in kind. The *passive* term is what diffusion does to any population at all: it smears cells over the landscape, erasing whatever advantage the release point happened to offer. It would be there for a dead cell. The *harvest* term is what the response actually earns—the mobility multiplied by the mean-square gradient the population currently sits in.

Result. *κ is a foraging efficiency*: nutrient collected per unit of squared gradient available. The same number that fixes where cells end up also fixes how well they eat on the way.

Two cancellations deserve names. The diffusivity shift δD_{eff} drops out of Eq. (17) entirely—spreading faster where food is plentiful does not, by itself, feed anyone—yet the level–jitter sensitivity $D_T^{(\eta)}$ survives inside κ. This is the exact mirror of Eq. (13), where it cancelled: *the rule that moves no centre of mass is precisely the one that feeds the population, and the rule that moves the centre of mass fastest need not feed it at all.*

The sign of the gain is the sign of κ, and it reads homely. Both $D_T^{(\eta)} < 0$ and $v^{(\eta)} < 0$ give $\kappa > 0$: a population profits by *slowing down and jittering less where the food is rich*—by lingering once it finds a good patch. That is area-restricted search, and Eq. (17) says exactly how much it is worth.

A plate assay reports neither a rate nor a displacement but a ratio, and on one mode of amplitude ε the zero-flux state gives it in closed form,

$$\frac{\rho_{\max}}{\rho_{\min}} = e^{2\mathrm{Pe}}, \qquad \mathrm{Pe} = \frac{\kappa\varepsilon}{D_{\text{eff}}^{(0)}}, \tag{18}$$

The **Péclet number** Pe is the only dimensionless group in the problem, and it is the number to quote when asking whether a mechanism matters. It is at once the logarithm of the fold-accumulation, the fractional density contrast a weak landscape produces, and—since drift overtakes diffusion at $\mathrm{Pe} \simeq 1$—the fractional reduction in the time a population needs to find a target. *A rule with $|\mathrm{Pe}| \ll 1$ is biologically invisible however impressive its microscopic sensitivity looks on paper.* Note again which coefficient appears: a steady state carries no flux, so δD_{eff} cancels and accumulation, like harvest, measures κ.

Finally, a timing point that matters for how long an assay should run. Spread and uptake read different parts of the landscape and so run on different clocks: the spread is set by the size of the arena, saturating after $\tau_R = 1/(D_{\text{eff}}^{(0)}k_1^2)$, while the uptake is locked to the food patches themselves, saturating after $\tau_\ell = 1/(D_{\text{eff}}^{(0)}q^2)$. Their ratio,

$$\frac{\tau_\ell}{\tau_R} = \left(\frac{k_1\ell}{2\pi\mathscr{R}}\right)^2 \simeq 0.086\left(\frac{\ell}{\mathscr{R}}\right)^2, \tag{19}$$

is small whenever the food is patchier than the dish; for the parameters below it is 1/186.

Result. *Foraging finishes long before mixing does.* A population collects essentially all the nutrient available to it while its density still looks far from steady. Local optimization is fast, global mixing is slow, and an assay that waits for the density to settle has waited a couple of hundred times longer than the biology required—and has spent that time measuring the dish rather than the cells.

Reading it off the data

uptake rate, once alignment is steady	*returns* κ, independently of the release
fold-accumulation $\rho_{\max}/\rho_{\min}$	*returns* κ once more, as $e^{2\mathrm{Pe}}$
cells accumulate but eat nothing extra	a rule with $\kappa = 0 \Rightarrow 2, 4$
$\lvert\mathrm{Pe}\rvert \ll 1$	the rule is biologically invisible, whatever its microscopic sensitivity
uptake plateaus, density still drifting	normal: the two run on clocks $(\ell/\mathscr{R})^2$ apart

IX. TRACKING: WHAT ONE CELL ADDS

Everything so far uses the density and polarity at a single time. Tracking individual cells costs more, and buys two things a population picture cannot supply: it separates rules the fields merge, and it pulls apart two notions that are constantly conflated—*navigating well* and *moving informatively.*

Watch one cell over a fixed *lag* τ, from t to $t+\tau$. Write $g = |\vec{\nabla}\eta|$ for the gradient strength, α for the *heading error*—the angle between where the cell points and where the food is—and $\mathcal{S}_t = (\vec{r}_t, \alpha_t)$ for the cell's state relative to the nutrient. Every rule is then judged by one and the same random variable, the *distance made good* up the gradient the cell actually faced,

$$Y^\tau = \big(\vec{r}_{t+\tau} - \vec{r}_t\big)\cdot\hat{g}(\vec{r}_t). \tag{20}$$

Two things can be asked of Y^τ. Its average is a *navigation velocity* $V_\parallel$: how fast this cell really climbs. Its full distribution is a *communication channel*, and the mutual information $\mathcal{I}$ it carries says how much an observer watching the cell learns about where it started relative to the food,

$$V_\parallel(t;\tau) = \frac{\mathbb{E}\big[Y^\tau\big]}{\tau}, \qquad \mathcal{I}(t;\tau) = \frac{I\big(\mathcal{S}_t; Y^\tau\big)}{\tau}, \tag{21}$$

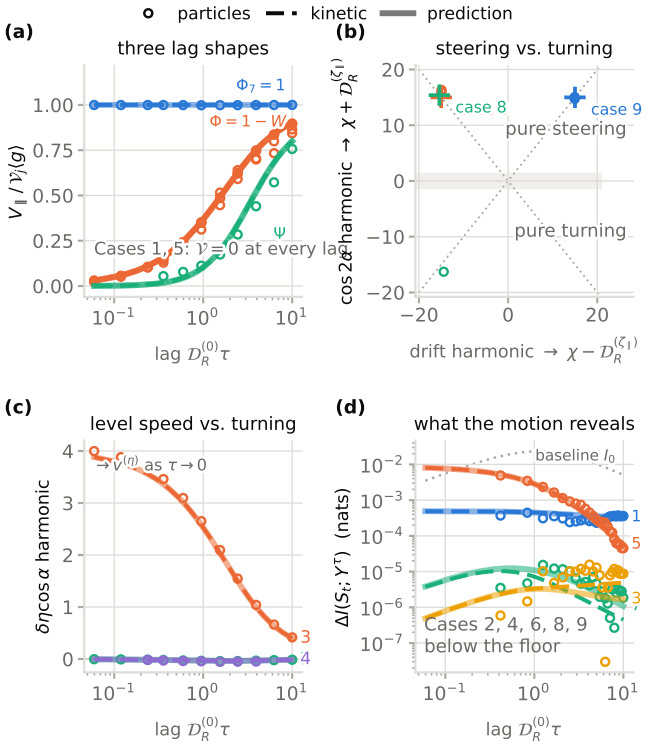


FIG. 5. **What tracking single cells adds, and what it does not.** All panels use a constant gradient and a common observation lag τ. (a) How fast a tracked cell really climbs, divided by the drift of the whole cloud. Every curve saturates at one—*a single cell watched long enough climbs at exactly the population's drift*—but reaches it through one of only three shapes, set by how many heading relaxations the rule needs before it can move a cell. The two jitter rules (1, 5) stay flat at zero at every lag. (b) Sorting the same tracks by initial heading returns two independent combinations, so pure steering (9) and pure turning (6, 8) separate—which no population measurement can do. (c) The same trick on the level rules separates level–speed (3), which returns the bare sensitivity $v^{(\eta)}$ as the lag closes, from level–wander and level–tumble (2, 4), which vanish with the lag. (d) How much a cell's motion reveals about its state, above the baseline that a merely swimming cell already provides. *Note the reversal:* the jitter rules, which never carry a cell up-gradient, dominate here, while the turning rules that navigate well sit below the estimator's floor.

Using the same variable for all nine rules is what puts a speed response and a tumble response on an equal footing: no rule-specific "controller action" is invented, and no coefficient is frozen during the window.

A. Navigation: the same drift, on a lag clock

Taking the release as the window start and working to first order gives, remarkably,

$$\boxed{V_{\parallel,j}(0;\tau) = \mathcal{V}_j\,\langle g\rangle\,\Phi_j\left(\mathcal{D}_R^{(0)}\tau\right)} \tag{22}$$

with $\mathcal{V}_j$ the very same pulse drift as Eq. (13), and only three lag shapes among all nine rules,

$$W(u) = \frac{1-e^{-u}}{u}, \qquad \Phi(u) = 1 - W(u) \simeq \frac{u}{2},$$
$$\Psi(u) = 1 - \frac{2-(2+u)e^{-u}}{u} \simeq \frac{u^2}{6}, \tag{23}$$

distributed as

$$\Phi_7 = 1, \quad \Phi_{3,6,8,9} = \Phi, \quad \Phi_{2,4} = \Psi, \quad \mathcal{V}_1 = \mathcal{V}_5 = 0. \tag{24}$$

Result. *Watched long enough, a tracked cell climbs at exactly the rate the whole cloud's centre of mass drifts*—individual navigation and pulse drift are one number. The lag decides only how much of that rate is visible, through one of just three curves.

Those three shapes are the short-time hierarchy again, now in the lag domain, and the counting is once more a counting of delays. Aimed–speed needs none: a cell that speeds up whenever it happens to point up-gradient makes progress on its first step. Level–speed, aimed–wander, aimed–tumble and steering need one—the heading must be biased before displacement follows. Level–wander and level–tumble need two. The jitter rules never navigate at any lag: a diffusivity that depends on position drifts the population but not the individual.

Sorting the windows by the cell's initial heading buys more. The average climb carries $\chi - \mathcal{D}_R^{(\zeta_\parallel)}$, but the $\cos 2\alpha$ harmonic of the same data carries $\chi + \mathcal{D}_R^{(\zeta_\parallel)}$. Two independent combinations separate genuine steering from heading-dependent turning—the one degeneracy no population measurement can break—and it is no accident that the missing information sits in exactly the harmonic the closure discarded.

B. Information: what the motion gives away

At short lag the channel is Gaussian with no approximation, $Y^\tau = \tau\mu(\mathcal{S}) + \sqrt{2D_T(\mathcal{S})\tau}\,Z$ with $\mu = v(\mathcal{S})\cos\alpha$, so the rate follows in closed form,

$$\boxed{\mathcal{J}(t;0^+) = \frac{\mathrm{Var}[\mu(\mathcal{S}_t)]}{4D_T^{(0)}} + \frac{\mathrm{Var}[\delta(\mathcal{S}_t)]}{4\tau}} \tag{25}$$

with δ the relative change in diffusivity, $D_T = D_T^{(0)}[1+\delta]$. Two features of this formula do the work.

First, the baseline is already positive: $\mathcal{J}_0 = (v^{(0)})^2/8D_T^{(0)}$ even with no sensing at all. A cell that merely swims moves further along its own heading than across it, so watching it tells you something about which way it pointed. *Movement information is therefore not, by itself, evidence of sensing*—an inference that has to be made against this baseline, not against zero.

Second, only the rules that change the *step* are legible: the two speed rules through $\mathrm{Var}[\mu]$ and the two jitter rules through $\mathrm{Var}[\delta]$ [Eq. (F10)]. All four turning rules and steering leave the rate exactly at baseline, because

they change the heading before anything has moved. The two jitter rules are singular as well as dominant—their channel is a scale mixture rather than a shifted Gaussian, so the total information saturates and the *rate* diverges as $\tau \to 0$, which is why a common positive lag is required for any fair comparison. And the drift never appears at all: shifting every outcome by the same amount tells an observer nothing about the state.

Result. *Navigating well and moving informatively are dissociated.* The two jitter rules never carry a cell up-gradient at any lag, yet they dominate what its motion reveals; level–wander and level–tumble navigate perfectly well and reveal almost nothing. The comfortable intuition that a better forager must be a more legible one gets no support here, and an inference method that assumes it will misread the mechanism.

Reading it off the data

climbing rate flat at zero, every lag	a jitter rule $\Rightarrow 1, 5$
climbing rate saturates at once	aimed–speed $\Rightarrow 7$
information rate diverges as $\tau \to 0$	a jitter rule again $\Rightarrow 1, 5$
information rate sits at the swimming baseline	a turning rule $\Rightarrow 2, 4, 6, 8, 9$
$\cos 2\alpha$ harmonic off the turning line	steering $\Rightarrow 9$, separated from 6 and 8
$\delta\eta \cos\alpha$ harmonic finite as $\tau \to 0$	level–speed $\Rightarrow 3$, separated from 2 and 4

X. NUMERICAL TEST

Two independent numerical descriptions test the construction: direct simulation of the microscopic jump–diffusion, Eq. (5), and a conservative finite-volume integration of the closed system, Eq. (10). All three calculations share one release, one nutrient field and one parameter set, and no transport coefficient is fitted. The landscape is a single cosine of wavenumber $q\mathscr{R} = 7.0156$ and amplitude $\varepsilon = 0.25$, so every closed form above applies verbatim; Appendix G explains that choice, the calibration, and the flux discretization a chemotactic measurement is sensitive to.

Figure 2 shows level–jitter (Case 1), the strictest sign check available: cells accumulate where they jiggle *least*, and a theory that mishandles the position-dependent diffusivity gets the sign backwards. The three methods agree to relative L^2 density errors of 7.2×10^{-3} and 1.0×10^{-2}, at cosine similarities 0.99998 and 0.99996.

The four release observables give the sharpest test, because simulated cells estimate them as plain averages—no binning, no bandwidth, no field to reconstruct. Figure 3 confirms three predictions at once: the uniform background reproduces the ballistic-to-diffusive crossover at $t\mathscr{D}_R^{(0)} = 1$; every response enters at the power of t Table II assigns it, with fitted exponents within a per cent; and the structural zeros hold exactly. The shape $\mathbf{T}$ is measured here for the first time, and its sharpest prediction holds: the two jitter rules return elongations consistent with zero ($z = 1.2$ and -0.3), while the other seven match theory to 1.3%.

Figure 4 tests the saturation and harvest laws. Reading the width and elongation of one settled cloud returns κ and $\mathscr{V}$ to within 2.5% for every rule that has them, and every feeding rule collapses onto $1 - e^{-t/\tau_\ell}$ with a gain independent of wavelength, as Eq. (18) requires. Figure 5 tests the tracked-cell predictions against a solution of the kinetic problem itself, since the coarse-grained fields cannot produce a two-time law: the lag shapes, the harmonic separations and the information baseline all come through, with the turning rules four orders of magnitude below the two jitter rules.

Residuals are of two known kinds, and both are understood. Simulated cells and the coarse-grained solver differ by about a per cent on the polarity and centroid and by up to seven on the shape—the ordering of exposure to the discarded nematic harmonic. The solver then departs from the first-order theory at second order in the response amplitude: halving every sensitivity halves the gap. Once a response is resolved to better than a per cent, the residual settles at the size of the neglected harmonic, $(v^{(0)}q/\mathscr{D}_{R,2}^{(0)})^2 = 3.8\%$ here—so this window is exactly where the closure's own error becomes the limiting systematic.

XI. DISCUSSION

Three practical messages come out of all this, and they are worth separating from the machinery that produced them.

A centroid velocity is not a chemotactic coefficient. The mobility κ, which fixes where a population ends up and how much it eats, is a different number from the drift $\mathscr{V}$ that its centre of mass displays. They can differ in magnitude, and for a speed rule they differ in sign. Both come off a single release—the width of the settled cloud gives one, its elongation the other—so there is no excuse for reporting only the drift, as chemotaxis assays routinely do.

When you look decides what you can see. Early on, all nine rules are distinguishable; over an intermediate window they sort into six groups; at steady state only three numbers survive. An assay run to equilibrium has thrown most of the information away, and has waited far longer than the cells needed: feeding runs on the landscape's clock, mixing on the dish's, and here those are a factor of 186 apart.

A better forager is not a more informative one. The rules whose motion reveals most about a cell's state are precisely the two that never carry it up-gradient, while the best navigators sit at the baseline that mere swimming already supplies. Any inference scheme assuming that legibility tracks performance will read the mechanism backwards.

Table IV collects the practical content of all three in one place: each row an experiment, what it returns, and which rules it separates.

One mode, or many? We have worked throughout with a single sinusoid, and that was a choice, not a restriction. To first order in the contrast the theory is linear in the landscape, so an arbitrary η can be built by superposing modes and the responses simply add. But the reading becomes harder, and it is worth seeing why. Each mode saturates on its own clock, $\tau_\ell(q) \propto q^{-2}$, and contributes to the final width and elongation with its own q-dependent weight: the mobility term scales as q^{-2} while the run-length correction of Eq. (E8) does not. A broadband landscape therefore returns not κ and $\mathscr{V}$ but weighted integrals of them over the spectrum, and recovering the coefficients means deconvolving a measured $\tilde{\eta}(\vec{q})$—an inversion that is well posed in principle and ill conditioned in practice, since the short-wavelength modes decay first and contribute least. The practical advice is the obvious one: if you can choose the landscape, choose one wavelength, and repeat at a second to separate the two q-scalings. If you cannot—in a natural, patchy environment—the same formulas still apply mode by mode, but expect the answer to be a spectrum-weighted average rather than a clean coefficient.

Three approximations delimit everything above: the linear response laws, the polar closure, and the weak-landscape and wall-free windows. Two pairs of rules stay merged throughout, and both are broken by the same missing measurement—nematic order—which would also resolve the few-per-cent closure systematic. That is the natural next experiment.

Appendix A: Model, kinetic equation and boundary condition

1. Dimensionless variables

Take $\mathscr{R}$ as the length scale and let the reference translational diffusivity $\mathscr{D}_T^{(0)} > 0$ set the time scale $\mathscr{R}^2/\mathscr{D}_T^{(0)}$. Then

$$\vec{r} = \frac{\vec{r}_{\mathrm{phys}}}{\mathscr{R}}, \qquad t = \frac{\mathscr{D}_T^{(0)} t_{\mathrm{phys}}}{\mathscr{R}^2}, \tag{A1}$$

and $\vec{r}$ lives in the unit disk $\mathbb{D} = \{\vec{r} \in \mathbb{R}^2 : |\vec{r}| < 1\}$. Everything below is dimensionless. Speeds are measured in $\mathscr{D}_T^{(0)}/\mathscr{R}$, translational diffusivities in $\mathscr{D}_T^{(0)}$, and angular diffusivities and tumbling rates in $\mathscr{D}_T^{(0)}/\mathscr{R}^2$, so that $D_T^{(0)} = 1$. We nonetheless keep writing $D_T^{(0)}$ in the formulas: it keeps their dimensional origin visible and lets them be read straight against simulation parameters. The cell number N is left unnormalized.

A cell is a point $\vec{r} \in \mathbb{D}$ carrying a heading angle $\theta \in [0, 2\pi)$ measured from the first coordinate axis. The heading and its perpendicular are

$$\begin{aligned} \hat{n}(\theta) &= (\cos\theta, \sin\theta)^{\mathsf{T}}, \\ \hat{n}_\perp(\theta) &= (-\sin\theta, \cos\theta)^{\mathsf{T}}, \end{aligned} \qquad \partial_\theta \hat{n} = \hat{n}_\perp, \quad \partial_\theta^2 \hat{n} = -\hat{n}. \tag{A2}$$

2. Admissibility of the linear response

Equation (3) is a linear response law, so it makes sense only where every speed, diffusivity and rate stays nonnegative. A sufficient condition is

$$|X^{(\eta)}|\, \|\delta\eta\|_\infty + |X^{(\zeta_\parallel)}|\, \|\vec{\nabla}\eta\|_\infty \le X^{(0)}, \tag{A3}$$

for each X, with $\|\cdot\|_\infty$ the largest magnitude over $\mathbb{D}$. This marks where the linearization holds; it is not a saturation law. Check the two terms separately, because a gentle nutrient contrast does not guarantee a gentle gradient. The Fokker–Planck problem below needs one thing more, $D_T > 0$, or the spatial equation degenerates. Steering is bounded on its own by $|\chi|\, \|\vec{\nabla}\eta\|_\infty / \mathscr{D}_R^{(0)} \ll 1$, with $\mathscr{D}_R^{(0)}$ the reference polarity-relaxation rate of Eq. (7).

3. The stochastic process and the wall

Here $\vec{W}_t$ is a two-dimensional standard Wiener process, W_t^θ an independent scalar one, and every coefficient is evaluated at the current state through Eq. (3). The pure-jump process $\mathscr{J}_t^{\mathrm{tum}}$ stays constant between tumbles; at a tumble it resets the heading from θ_{t^-} to θ_{t^+} and leaves the position alone.

Two short-time relations fix the normalization of the diffusivities. Hold the coefficients fixed over Δt: the translational noise then produces a mean-square displacement $4D_T\Delta t$, and the rotational noise an angular variance $2D_R\Delta t$.

The wall stops cells without turning them: a displacement that would leave $\mathbb{D}$ loses its outward normal component and the cell slides along the boundary on what is left, heading untouched. This normal-constraint (Skorokhod) reflection becomes the no-flux condition Eq. (A9) in phase space, imposed for every heading separately. Because the heading is untouched, a cell pressed into the wall stays pressed until it turns away, and a polarity layer builds along the boundary. Every result below lives inside the wall-free window of Sec. IV, where no cell reaches the wall at all, so we state the rule and then never use it.

4. Tumble rate and turning law

Tumbling needs two ingredients: the rate $\lambda(\vec{r}, \theta)$ says when a cell tumbles, and the normalized kernel $\mathscr{T}(\theta' \mid \theta)$ says where it points afterwards. Both are evaluated at the pre-tumble state. For a rotationally invariant, achiral kernel,

$$\begin{gathered} \mathscr{T}(\theta' \mid \theta) = \widehat{\mathscr{T}}(\Delta\theta), \qquad \widehat{\mathscr{T}}(\Delta\theta) = \widehat{\mathscr{T}}(-\Delta\theta), \\ \int_0^{2\pi} \mathscr{T}(\theta' \mid \theta)\, \mathrm{d}\theta' = 1, \end{gathered} \tag{A4}$$

where $\Delta\theta = \theta' - \theta \pmod{2\pi}$. The one number we need from this kernel is its first harmonic,

$$\mathscr{P}_{\text{turn}} = \int_0^{2\pi} \cos(\Delta\theta)\, \widehat{\mathscr{T}}\,(\Delta\theta)\, \mathrm{d}(\Delta\theta), \quad -1 \le \mathscr{P}_{\text{turn}} \le 1, \tag{A5}$$

the mean heading a tumble retains: $\mathscr{P}_{\text{turn}} = 1$, 0 and -1 are a forward run, a uniform reorientation and a U-turn. Achirality kills the transverse mean, so a tumble does nothing but damp alignment, contributing $\lambda^{(0)}(1-\mathscr{P}_{\text{turn}})$ to the polarity-relaxation rate below.

5. Fokker–Planck master equation

Let $P(\vec{r}, \theta, t)$ be the number density on the phase space $\mathbb{D} \times \mathbb{S}^1$, so that

$$\int_{\mathbb{D}} \int_0^{2\pi} P(\vec{r}, \theta, t)\, \mathrm{d}\theta\, \mathrm{d}^2 r = N, \tag{A6}$$

with initial data $P(\vec{r}, \theta, 0) = P_0$ nonnegative, normalized to N, 2π-periodic, and compatible with no flux at the wall.

The Itô process of Eq. (5) carries spatial and angular currents

$$\vec{J}_{\text{sp}} = v\hat{n}P - \vec{\nabla}(D_T P), \qquad J_{\text{ang}} = \chi\zeta_\perp P - \partial_\theta(D_R P), \tag{A7}$$

and the Fokker–Planck master equation

$$\begin{aligned} \partial_t P = & -\vec{\nabla}\cdot\vec{J}_{\text{sp}} - \partial_\theta J_{\text{ang}} \\ & + \int_0^{2\pi} \lambda(\vec{r}, \theta')\mathscr{T}\,(\theta \mid \theta')P(\vec{r}, \theta', t)\, \mathrm{d}\theta' \\ & - \lambda(\vec{r}, \theta)P(\vec{r}, \theta, t). \end{aligned} \tag{A8}$$

The gain term uses the source rate $\lambda(\vec{r}, \theta')$, and normalization of T balances gain against loss. One detail deserves a flag now, because it decides signs later: the spatial diffusive current is $-\vec{\nabla}(D_T P)$, not $-D_T\vec{\nabla}P$. The difference is invisible while D_T is constant and decisive the moment the nutrient makes it vary.

Two boundary conditions close the problem. The wall is impermeable, so the normal spatial current vanishes for every heading,

$$\hat{\nu}\cdot\vec{J}_{\text{sp}} = 0 \qquad \vec{r} \in \partial\mathbb{D}, \quad \theta \in [0, 2\pi), \tag{A9}$$

with $\hat{\nu}$ the outward unit normal. The heading is an angle, so P and J_{ang} are 2π-periodic in θ. Integrating Eq. (A8) over the whole phase space then kills all three contributions in turn—the wall term by Eq. (A9), the angular term by periodicity, the tumble term by normalization of $\mathscr{T}$—and leaves N exactly conserved.

Appendix B: The polar closure and its validity

The exact density flux, before any closure, is

$$\begin{aligned} \vec{J}^{(\rho)} = & (v^{(0)} + v^{(\eta)}\delta\eta)\vec{p} + v^{(\zeta_\parallel)}\left(\tfrac{1}{2}\rho\mathbf{I} + \mathbf{Q}\right)\vec{\nabla}\eta \\ & - \vec{\nabla}\left[(D_T^{(0)} + D_T^{(\eta)}\delta\eta)\rho + D_T^{(\zeta_\parallel)}\vec{p}\cdot\vec{\nabla}\eta\right]. \end{aligned} \tag{B1}$$

Setting $\mathbf{Q} = \mathbf{0}$ requires three conditions. First, the population must stay near-isotropic,

$$\frac{2|\vec{p}|}{\rho} \ll 1. \tag{B2}$$

Second, on the scale L_η over which the landscape varies, transport must be slow compared with nematic decay,

$$\frac{v^{(0)}}{\mathscr{D}_{R,2}^{(0)} L_\eta} \ll 1. \tag{B3}$$

Third, Case 7 needs a condition of its own. Take the traceless second angular moment of Eq. (A8) and keep the isotropic part of the fourth moment. At $\vec{p} = \vec{0}$ exactly one nutrient source of $\mathbf{Q}$ survives, the Case-7 term

$$\Sigma_{ij} = -\frac{v^{(\zeta_\parallel)}}{8}\Big[\partial_i(\rho\,\partial_j\eta) + \partial_j(\rho\,\partial_i\eta) - \delta_{ij}\vec{\nabla}\cdot(\rho\vec{\nabla}\eta)\Big]; \tag{B4}$$

where every other rule reaches $\mathbf{Q}$ only through $\vec{p}$. Balancing Σ_{ij} against relaxation at $\mathscr{D}_{R,2}^{(0)}$ and imposing $2|\mathbf{Q}|/\rho \ll 1$ gives

$$\frac{|v^{(\zeta_\parallel)}|}{\mathscr{D}_{R,2}^{(0)}}\left(\|\nabla\nabla\eta\|_\infty + \frac{\|\vec{\nabla}\eta\|_\infty}{L_\rho}\right) \ll 1, \tag{B5}$$

with $\|\nabla\nabla\eta\|_\infty$ the largest Hessian magnitude over the sampled region and L_ρ the scale on which the density varies. Both terms matter: expanding $\partial_i(\rho\partial_j\eta)$ gives $\rho\,\partial_i\partial_j\eta$ and $\partial_i\rho\,\partial_j\eta$, and whenever the density varies on the landscape's own scale the second is as large as the first.

The three constitutive pieces of Eqs. (10) are

$$\begin{aligned} \vec{J}^{(\rho)} = & (v^{(0)} + v^{(\eta)}\delta\eta)\vec{p} + \frac{v^{(\zeta_\parallel)}\rho}{2}\vec{\nabla}\eta \\ & - \vec{\nabla}\left[(D_T^{(0)} + D_T^{(\eta)}\delta\eta)\rho + D_T^{(\zeta_\parallel)}\vec{p}\cdot\vec{\nabla}\eta\right], \end{aligned} \tag{B6}$$

$$\begin{aligned} J_{ij}^{(p)} = & \frac{\rho}{2}(v^{(0)} + v^{(\eta)}\delta\eta)\delta_{ij} \\ & + \frac{v^{(\zeta_\parallel)}}{4}\left[p_i\partial_j\eta + p_j\partial_i\eta + (\vec{p}\cdot\vec{\nabla}\eta)\delta_{ij}\right] \\ & - \partial_j\left[(D_T^{(0)} + D_T^{(\eta)}\delta\eta)p_i + \frac{D_T^{(\zeta_\parallel)}\rho}{2}\partial_i\eta\right], \end{aligned} \tag{B7}$$

$$\vec{R}^{(p)} = -\left[\mathscr{D}_R^{(0)} + \mathscr{D}_R^{(\eta)}\delta\eta\right]\vec{p} + \frac{(\chi - \mathscr{D}_R^{(\zeta_\parallel)})\rho}{2}\vec{\nabla}\eta, \tag{B8}$$

Appendix C: Uniform background and the weak landscape

1. Wall-free window and the plane limit

On the plane the system is linear and translationally invariant, so it has a Green propagator: the density and polarity produced at $(\vec{r}, t)$ by a point release at $(\vec{r}', t')$, depending only on $\vec{r} - \vec{r}'$ and $t - t'$.

That one object does two jobs. Convolved with an arbitrary initial profile it gives the uniform-background evolution. Convolved instead—through Duhamel's principle—with the forcing a weak landscape spreads through space, it gives the first-order population response.

2. Spectrum and Green propagator

For $\eta = \eta^{(0)}$ ($\delta\eta = 0$, $\vec{\nabla}\eta = \vec{0}$), Eqs. (10) collapse to

$$\partial_t \rho = D_T^{(0)} \nabla^2 \rho - v^{(0)} \vec{\nabla} \cdot \vec{p}, \tag{C1a}$$

$$\partial_t \vec{p} = D_T^{(0)} \nabla^2 \vec{p} - \mathscr{D}_R^{(0)} \vec{p} - \frac{v^{(0)}}{2} \vec{\nabla} \rho. \tag{C1b}$$

Read these two lines physically: translational diffusion acts equally on both fields, the speed $v^{(0)}$ couples density gradients to polarity flux and back, and alignment relaxes isotropically at $\mathscr{D}_R^{(0)}$. With the convention $\tilde{f}(\vec{k}) = \int e^{-i\vec{k}\cdot\vec{r}} f \, \mathrm{d}^2 r$, so that $\vec{\nabla} \to i\vec{k}$ and $\nabla^2 \to -k^2$, decompose the polarity relative to $\hat{\vec{k}} = \vec{k}/k$,

$$\tilde{\vec{p}} = \tilde{p}_L \, \hat{\vec{k}} + \tilde{p}_T \, \hat{\vec{k}}_\perp, \qquad \hat{\vec{k}}_\perp = (-\hat{k}_2, \hat{k}_1)^{\mathsf{T}}. \tag{C2}$$

Gradients and divergences act strictly along $\vec{k}$, so the transverse polarity generates no spatial flux and feels no density coupling. It simply decays,

$$\tilde{p}_T(\vec{k}, t) = \tilde{p}_{T,0}(\vec{k}) \, e^{-\left(D_T^{(0)} k^2 + \mathscr{D}_R^{(0)}\right) t}. \tag{C3}$$

Density and longitudinal polarity form a coupled pair, $\partial_t (\tilde{\rho}, \tilde{p}_L)^{\mathsf{T}} = \mathbf{M}(k) (\tilde{\rho}, \tilde{p}_L)^{\mathsf{T}}$, with

$$\mathbf{M}(k) = \begin{pmatrix} -D_T^{(0)} k^2 & -i v^{(0)} k \\ -\frac{i v^{(0)} k}{2} & -\left(D_T^{(0)} k^2 + \mathscr{D}_R^{(0)}\right) \end{pmatrix}. \tag{C4}$$

Define the mean damping and spectral splitting

$$\mu_k = \tfrac{1}{2} \operatorname{Tr}(-\mathbf{M}) = D_T^{(0)} k^2 + \tfrac{1}{2} \mathscr{D}_R^{(0)} \tag{C5}$$

and the traceless remainder $\mathbf{A}(k)$, whose square is a multiple of the identity, $\mathbf{A}^2 = (\sigma_k/2)^2 \mathbf{I}$, with the spectral splitting

$$\sigma_k = \sqrt{\left[\mathscr{D}_R^{(0)}\right]^2 - 2\left[v^{(0)}\right]^2 k^2}. \tag{C6}$$

The longitudinal propagator $\mathbf{G}_L(k,t) = e^{\mathbf{M}(k) t}$ is

$$\mathbf{G}_L(k,t) = e^{-\mu_k t} \begin{pmatrix} \cosh\left(\frac{\sigma_k t}{2}\right) + \frac{\mathscr{D}_R^{(0)}}{\sigma_k} \sinh\left(\frac{\sigma_k t}{2}\right) & -\frac{2 i v^{(0)} k}{\sigma_k} \sinh\left(\frac{\sigma_k t}{2}\right) \\ -\frac{i v^{(0)} k}{\sigma_k} \sinh\left(\frac{\sigma_k t}{2}\right) & \cosh\left(\frac{\sigma_k t}{2}\right) - \frac{\mathscr{D}_R^{(0)}}{\sigma_k} \sinh\left(\frac{\sigma_k t}{2}\right) \end{pmatrix}, \tag{C7}$$

acting on any initial $(\tilde{\rho}_0, \tilde{p}_{L,0})^{\mathsf{T}}$. Together with Eq. (C3) this is the complete uniform-background dynamics—no approximation, no fitting.

The sign of σ_k^2 splits the behaviour into two regimes. For $k < k_c = \mathscr{D}_R^{(0)}/(\sqrt{2} v^{(0)})$ the splitting σ_k is real and the longitudinal modes decay at the two rates $\mu_k \mp \sigma_k/2$, which as $k \to 0$ become

$$s_{\mathrm{slow}}(k) = \left(D_T^{(0)} + \frac{(v^{(0)})^2}{2\mathscr{D}_R^{(0)}}\right) k^2 + \mathcal{O}(k^4), \tag{C8}$$

$$s_{\mathrm{fast}}(k) = \mathscr{D}_R^{(0)} + \left(D_T^{(0)} - \frac{(v^{(0)})^2}{2\mathscr{D}_R^{(0)}}\right) k^2 + \mathcal{O}(k^4). \tag{C9}$$

On macroscopic scales density therefore decays diffusively with the enhanced diffusivity Eq. (C22) the familiar enhancement of diffusion by self-propulsion, while polarity mismatches relax at $\mathscr{D}_R^{(0)}$.

Above k_c the splitting turns imaginary and the hyperbolic functions become trigonometric, $\cosh(\sigma_k t/2) \to \cos(|\sigma_k| t/2)$ and $\sigma_k^{-1} \sinh(\sigma_k t/2) \to |\sigma_k|^{-1} \sin(|\sigma_k| t/2)$. Spatial gradients then drive transient polar-advection waves at frequency $\omega_k = |\sigma_k|/2$ under the envelope $e^{-\mu_k t}$. The wavenumber k_c is where long-range macro-diffusion gives way to short-range transient waves: cells cross a wavelength before they forget their heading.

3. Point release

For a centered release of N cells with isotropic headings, $P(\vec{r}, \theta, 0) = (N/2\pi)\delta^{(2)}(\vec{r})$, i.e. $\rho_0 = N\delta^{(2)}(\vec{r})$ and $\vec{p}_0 = \vec{0}$, the Fourier initial data are $\tilde{\rho}_0 = N$, $\tilde{p}_{L,0} = 0$, and Eq. (C7) gives

$$\tilde{\rho}^{(0)}(k,t) = N A_k(t), \qquad \tilde{p}_i^{(0)}(\vec{k}, t) = -iN\hat{k}_i \, B_k(t), \tag{C10}$$

with the per-cell Green amplitudes

$$A_k(t) = e^{-\mu_k t} \left[\cosh\left(\frac{\sigma_k t}{2}\right) + \frac{\mathscr{D}_R^{(0)}}{\sigma_k} \sinh\left(\frac{\sigma_k t}{2}\right)\right], \tag{C11a}$$

$$B_k(t) = e^{-\mu_k t} \frac{v^{(0)} k}{\sigma_k} \sinh\left(\frac{\sigma_k t}{2}\right). \tag{C11b}$$

Both amplitudes have plain readings. A_k follows the pulse as it broadens, carrying it from ballistic spreading at short times to isotropic diffusion at long ones. B_k follows the radial polarity that builds up while cells stream away from the release point, and that fades once their headings decorrelate.

The source is isotropic, so the real-space fields follow from inverse Hankel transforms,

$$\rho^{(0)}(r,t) = \frac{N}{2\pi}\int_0^\infty k\,\mathrm{d}k\,J_0(kr)A_k(t), \tag{C12a}$$

$$p_r^{(0)}(r,t) = \frac{N}{2\pi}\int_0^\infty k\,\mathrm{d}k\,J_1(kr)B_k(t). \tag{C12b}$$

Equations (C11) and (C12) carry single-cell dynamics straight onto the population propagator. Five single-cell numbers, $(v^{(0)}, D_T^{(0)}, D_R^{(0)}, \lambda^{(0)}, \mathscr{P}_{\mathrm{turn}})$, fix the entire spatial evolution. Nothing is left to fit.

4. Duhamel expansion and the first-order kernels

Let the landscape be weak and static,

$$\eta(\vec{r}) = \eta^{(0)} + \varepsilon\eta_1(\vec{r}), \qquad 0 < \varepsilon \ll 1, \tag{C13}$$

and expand $\rho = \rho^{(0)} + \varepsilon\rho^{(1)} + \mathcal{O}(\varepsilon^2)$ and $\vec{p} = \vec{p}^{(0)} + \varepsilon\vec{p}^{(1)} + \mathcal{O}(\varepsilon^2)$. The initial data stay fixed, so the corrections start from nothing, $\rho^{(1)}(\vec{r},0) = 0$ and $\vec{p}^{(1)}(\vec{r},0) = \vec{0}$, and $(\rho^{(0)}, \vec{p}^{(0)})$ is exactly the uniform-background evolution of App. C.

At $\mathcal{O}(\varepsilon)$ the coupling is a convolution: a nutrient mode at $\vec{k}'$ meets a population mode at $\vec{k}-\vec{k}'$ and produces a response at $\vec{k}$. Three kernels carry it—one for the density flux, one for the polarity flux, one for the angular source—and all nine microscopic rules live in them, and nowhere else. Their explicit forms, and the effective forcings $\tilde{f}_\rho$ and $\tilde{f}_{p,i}$ obtained by integrating them against the nutrient spectrum $\tilde{\eta}_1(\vec{k}')$, are collected in Eqs. (C14)–(C18) below. What matters here is their structure, and it is the structure that drives everything below: a response acts either on $\tilde{\rho}^{(0)}$ or on $\tilde{\vec{p}}^{(0)}$, and it enters either a flux or the polarity source.

Explicitly,

$$\tilde{J}^{(1)}_{\rho,i}(\vec{k},\vec{k}',t') = v^{(\eta)}\tilde{p}_i^{(0)} + \frac{iv^{(\zeta_\parallel)}}{2}\tilde{\rho}^{(0)}k_i' - iD_T^{(\eta)}\tilde{\rho}^{(0)}k_i + D_T^{(\zeta_\parallel)}\left[\tilde{\vec{p}}^{(0)}\cdot\vec{k}'\right]k_i, \tag{C14}$$

$$\tilde{J}^{(p)(1)}_{ij}(\vec{k},\vec{k}',t') = \frac{v^{(\eta)}}{2}\tilde{\rho}^{(0)}\delta_{ij} + \frac{iv^{(\zeta_\parallel)}}{4}\left[\tilde{p}_i^{(0)}k_j' + k_i'\tilde{p}_j^{(0)} + \left(\tilde{\vec{p}}^{(0)}\cdot\vec{k}'\right)\delta_{ij}\right] - iD_T^{(\eta)}\tilde{p}_i^{(0)}k_j + \frac{D_T^{(\zeta_\parallel)}}{2}\tilde{\rho}^{(0)}k_i'k_j, \tag{C15}$$

$$\tilde{R}^{(1)}_{p,i}(\vec{k},\vec{k}',t') = -\mathscr{D}_R^{(\eta)}\tilde{p}_i^{(0)} + \frac{i(\chi - \mathscr{D}_R^{(\zeta_\parallel)})}{2}\tilde{\rho}^{(0)}k_i', \tag{C16}$$

with every zeroth-order field evaluated at $(\vec{k}-\vec{k}',t')$. Integrating them against the nutrient spectrum gives the effective forcings

$$\tilde{f}_\rho(\vec{k},t') = \int_{\mathbb{R}^2}\frac{\mathrm{d}^2k'}{(2\pi)^2}\tilde{\eta}_1(\vec{k}')\left[-ik_i\tilde{J}^{(1)}_{\rho,i}(\vec{k},\vec{k}',t')\right], \tag{C17}$$

$$\tilde{f}_{p,i}(\vec{k},t') = \int_{\mathbb{R}^2}\frac{\mathrm{d}^2k'}{(2\pi)^2}\tilde{\eta}_1(\vec{k}')\left[-i\tilde{J}^{(p)(1)}_{ij}(\vec{k},\vec{k}',t')k_j + \tilde{R}^{(1)}_{p,i}(\vec{k},\vec{k}',t')\right], \tag{C18}$$

which enter Eq. (C19).

Write $\tilde{f}_L = \hat{k}_i\tilde{f}_{p,i}$ and $\tilde{f}_T = \hat{k}_{\perp,i}\tilde{f}_{p,i}$ for the longitudinal and transverse projections of the polarity forcing. By Duhamel's principle the first-order fields are then the convolution of that forcing with the unperturbed propagator,

$$\begin{pmatrix}\tilde{\rho}^{(1)}\\ \tilde{p}_L^{(1)}\end{pmatrix}(\vec{k},t) = \int_0^t \mathbf{G}_L(k,t-t')\begin{pmatrix}\tilde{f}_\rho\\ \tilde{f}_L\end{pmatrix}(\vec{k},t')\,\mathrm{d}t', \tag{C19a}$$

$$\tilde{p}_T^{(1)}(\vec{k},t) = \int_0^t e^{-\left[D_T^{(0)}k^2 + \mathscr{D}_R^{(0)}\right](t-t')}\tilde{f}_T(\vec{k},t')\,\mathrm{d}t'. \tag{C19b}$$

Equation (C19) completes the wall-free solution. Every correction is an explicit convolution with the measured nutrient spectrum $\tilde{\eta}_1(\vec{k}')$: no mode of the landscape has been coarse-grained away, and a landscape given as data can be used as data. Specializing to the point release of Sec. C 3 costs one substitution,

$$\tilde{\rho}^{(0)} = NA_{|\Delta\vec{k}|}(t'), \quad \tilde{p}_i^{(0)} = -iN\frac{\Delta k_i}{|\Delta\vec{k}|}B_{|\Delta\vec{k}|}(t'), \tag{C20}$$

with $\Delta\vec{k} = \vec{k} - \vec{k}'$, into the kernels. The physical fields then follow from two-dimensional inverse transforms of $\tilde{\rho}$ and $\tilde{p}_i$: once the landscape breaks rotational symmetry the Hankel reduction no longer applies. At zero wavevector, conservation and Eq. (10b) give $\tilde{\rho}(\vec{0},t) = N$ and $\tilde{p}_i(\vec{0},t) = \varepsilon \int_0^t e^{-\mathscr{D}_R^{(0)}(t-t')} \tilde{f}_{p,i}(\vec{0},t')\,\mathrm{d}t' + \mathcal{O}(\varepsilon^2)$.

5. Counting the transient exponents

The Duhamel integral has a second use. Read at short times, it sorts the nine rules by how far each one sits from the observable, and the counting is easy to follow.

A release starts with density but no polarity: $\tilde{\rho}^{(0)} = N + \mathcal{O}(t)$ and $\tilde{p}^{(0)} = \mathcal{O}(t)$ [Eq. (C11)], and the off-diagonal blocks of $\mathbf{G}_L$ are themselves $\mathcal{O}(t)$. So a response term pays one power of t for the Duhamel integration itself, a second if it multiplies $\vec{p}^{(0)}$ rather than $\rho^{(0)}$, and a third if it enters the polarity equation rather than the density flux. The tally is

$$\begin{aligned} &\delta\rho \sim t: \ \{1,7\}; \quad t^2: \ \{3,5,6,8,9\}; \quad t^3: \ \{2,4\}, \\ &\delta\vec{p} \sim t: \ \{3,5,6,8,9\}; \quad t^2: \ \{1,2,4,7\}. \end{aligned} \tag{C21}$$

Only two rules move density at first order in time, and they are the two that act on the density flux through ρ itself: $D_T^{(\eta)}$ and $v^{(\zeta_\parallel)}$. Cases 2 and 4 sit farthest away. They modulate the decay of a polarity that must be built before it can decay, and each of those steps costs a power of t.

This is a statement about experimental design, not only about asymptotics. Over a window $t \ll 1/\mathscr{D}_R^{(0)}$ the Case-2 density response stands to the Case-1 response as $\mathscr{D}_R^{(\eta)}(v^{(0)})^2 t^2 / D_T^{(\eta)}$, up to a numerical factor; for the parameters of Sec. X that ratio is 1.2×10^{-2}.

A release watched for a fifth of an orientational relaxation time therefore says almost nothing about how the nutrient level shifts the turning statistics—however many cells are tracked. Seeing Cases 2 and 4 in density needs $t \gtrsim 1/\mathscr{D}_R^{(0)}$, which is precisely where the fast mode of Eq. (C9) has died and the long-time reduction below takes over. Section X bears this out: at unchanged ensemble size the two carry a thirtieth of the Case-1 density response at $t\mathscr{D}_R^{(0)} \simeq 0.36$ and more than half of it at $t\mathscr{D}_R^{(0)} \simeq 1.78$. Their polarity signature arrives one power of t sooner, and is the better thing to watch early.

The enhanced diffusivity of the uniform background is

$$\begin{aligned} D_{\mathrm{eff}}^{(0)} &= D_T^{(0)} + \frac{(v^{(0)})^2}{2\mathscr{D}_R^{(0)}} \\ &= D_T^{(0)} + \frac{(v^{(0)})^2}{2\left[D_R^{(0)} + \lambda^{(0)}(1-\mathscr{P}_{\mathrm{turn}})\right]}, \end{aligned} \tag{C22}$$

and the landscape shifts it by

$$\delta D_{\mathrm{eff}}(\vec{r}) = \left[D_T^{(\eta)} + \frac{v^{(0)} v^{(\eta)}}{\mathscr{D}_R^{(0)}} - \frac{(v^{(0)})^2 \mathscr{D}_R^{(\eta)}}{2[\mathscr{D}_R^{(0)}]^2} \right] \eta_1(\vec{r}), \tag{C23}$$

TABLE III. **What survives at long times—and what is lost.** After many heading relaxations the nine rules compress onto only three measurable numbers: a shift in how fast headings relax, a shift in the spreading rate δD_{eff}, and the mobility κ that sets accumulation. A dash means no leading-order contribution. *Read the columns for the degeneracies*: level–wander and level–tumble (2, 4) enter identically, the three aimed-turning rules (6, 8, 9) enter only through κ, and aimed–jitter (5) vanishes from all three—so waiting for steady state destroys most of the information. Here $\bar{P} = 1 - \mathscr{P}_{\mathrm{turn}}$, and the pulse drift is $\mathscr{V} = \kappa + \delta D'_{\mathrm{eff}}$.

Case	Sensitivity	$1/\tau_{\mathrm{fast}} - \mathscr{D}_R^{(0)}$	δD_{eff}	κ
1	$D_T^{(\eta)}$	—	$D_T^{(\eta)}$	$-D_T^{(\eta)}$
2	$D_R^{(\eta)}$	$D_R^{(\eta)}$	$-\frac{(v^{(0)})^2 D_R^{(\eta)}}{2[\mathscr{D}_R^{(0)}]^2}$	—
3	$v^{(\eta)}$	—	$\frac{v^{(0)} v^{(\eta)}}{\mathscr{D}_R^{(0)}}$	$-\frac{v^{(0)} v^{(\eta)}}{2\mathscr{D}_R^{(0)}}$
4	$\lambda^{(\eta)}$	$\bar{P}\lambda^{(\eta)}$	$-\frac{(v^{(0)})^2 \bar{P}\lambda^{(\eta)}}{2[\mathscr{D}_R^{(0)}]^2}$	—
5	$D_T^{(\zeta_\parallel)}$	—	—	—
6	$D_R^{(\zeta_\parallel)}$	—	—	$-\frac{v^{(0)} D_R^{(\zeta_\parallel)}}{2\mathscr{D}_R^{(0)}}$
7	$v^{(\zeta_\parallel)}$	—	—	$\frac{v^{(\zeta_\parallel)}}{2}$
8	$\lambda^{(\zeta_\parallel)}$	—	—	$-\frac{v^{(0)} \bar{P}\lambda^{(\zeta_\parallel)}}{2\mathscr{D}_R^{(0)}}$
9	χ	—	—	$\frac{v^{(0)}\chi}{2\mathscr{D}_R^{(0)}}$

Appendix D: Long-time reduction and mode inversion

At long times ($t \gg 1/\mathscr{D}_R^{(0)}$) and long wavelengths the fast polarity mode has died and only the slow branch is left. Polarity then follows the density instead of leading it: setting $\partial_t \vec{p} = \vec{0}$ in Eq. (10b) and keeping the leading spatial gradients gives the adiabatic polarity

$$\vec{p} = \frac{1}{\mathscr{D}_R} \left[-\vec{\nabla}\left(\frac{\rho v}{2}\right) + \left(\chi - \mathscr{D}_R^{(\zeta_\parallel)}\right)\frac{\rho}{2}\vec{\nabla}\eta \right], \tag{D1}$$

in which $v = v^{(0)} + v^{(\eta)}\delta\eta$ and $\mathscr{D}_R = \mathscr{D}_R^{(0)} + \mathscr{D}_R^{(\eta)}\delta\eta$ still carry their landscape dependence. Substituting this into Eq. (B6) collapses the density dynamics to ordinary drift–diffusion,

$$\partial_t \rho = \vec{\nabla} \cdot \left[D_{\mathrm{eff}}(\vec{r})\vec{\nabla}\rho - \varepsilon\,\kappa\,\rho\,\vec{\nabla}\eta_1 \right], \tag{D2}$$

with a locally shifted diffusivity $D_{\mathrm{eff}}(\vec{r}) = D_{\mathrm{eff}}^{(0)} + \varepsilon\,\delta D_{\mathrm{eff}}(\vec{r})$, Eq. (C23) and a single chemotactic mobility Eq. (12) This is the leading term of a joint expansion in ε and in spatial gradients; every term it keeps carries two spatial derivatives in all. Case 5 first contributes at fourth order, through $-\varepsilon D_T^{(\zeta_\parallel)} v^{(0)} \nabla^2(\vec{\nabla}\rho \cdot \vec{\nabla}\eta_1)/2\mathscr{D}_R^{(0)}$, which is why $D_T^{(\zeta_\parallel)}$ appears in none of the coefficients below. It will reappear in Sec. VII.

The second bracket of Eq. (12) is easy to miss and not optional. A level response makes the transport coefficients themselves functions of position, and a spatially varying diffusivity or speed moves density with no directional

sensing whatever. In the Itô convention the flux $-\vec{\nabla}(D_T\rho)$ contains the drift $-\rho\vec{\nabla}D_T$, and eliminating $\vec{p}$ adiabatically supplies the companion term $-\rho\vec{\nabla}(v^2)/(4\mathscr{D}_R^{(0)})$. With translational diffusion negligible this is the classical run–tumble accumulation $\rho \propto 1/v$ of a population whose speed depends on position [2, 3]: cells linger where they are slow.

In a bounded domain the zero-flux state of Eq. (D2) is $\rho \propto \exp[\varepsilon\kappa\eta_1/D_{\rm eff}^{(0)}]$ to $\mathcal{O}(\varepsilon)$. So κ is the mobility that decides where the population ends up.

One caution before reading observables off Eq. (D2): its coefficients are not constant. $D_{\rm eff}$ varies at the same order ε as the drift, so freezing it is not innocent.

The safe route is through the two lowest moments, which are exact to $\mathcal{O}(\varepsilon)$ and are what a release experiment actually reports. Writing $\vec{x}$ for the displacement from the release point and $\langle\cdot\rangle$ for the density-weighted average,

$$\begin{aligned} \frac{\mathrm{d}\langle\vec{x}\rangle}{\mathrm{d}t} &= \varepsilon\,\mathscr{V}\,\langle\vec{\nabla}\eta_1\rangle, \qquad \mathscr{V} = \kappa + \delta D'_{\rm eff}, \\ \frac{\mathrm{d}}{\mathrm{d}t}\big\langle|\vec{x}-\langle\vec{x}\rangle|^2\big\rangle &= 4\langle D_{\rm eff}\rangle + \mathcal{O}(\varepsilon^2), \end{aligned} \tag{D3}$$

where $\delta D'_{\rm eff}$ is the bracket of Eq. (C23), so that $\delta D_{\rm eff} = \delta D'_{\rm eff}\,\eta_1$.

The centroid velocity is therefore *not* $\kappa\vec{\nabla}\eta_1$. A spatially varying diffusivity displaces the centroid on its own, through $\langle\vec{\nabla}D_{\rm eff}\rangle$, and that displacement simply adds. Explicitly, Eq. (13) in which $D_T^{(\eta)}$ has cancelled exactly. A pulse spreading in a linearly varying Itô diffusivity does not move its centroid—even though that same response is what sets the accumulation.

The distinction is physical, not bookkeeping, and the two coefficients can even come with opposite signs. Take a pure speed response (Case 3) with $v^{(\eta)} > 0$, so that $\kappa < 0$ while $\mathscr{V} > 0$. Early on the pulse centroid moves *toward* high nutrient, simply because cells spread faster on that side. Late on the population piles up where cells are slow, as $\rho \propto 1/v$ demands. One rule, two directions, two different questions.

Three structural facts follow. All three level responses set the spreading rate, and two of them, $D_T^{(\eta)}$ and $v^{(\eta)}$, also set part of the mobility κ—with weights that differ from their weights in $\delta D_{\rm eff}$. The gradient responses $(v^{(\zeta_\parallel)}, \mathscr{D}_R^{(\zeta_\parallel)}, \chi)$ enter κ alone. And Case 5, $D_T^{(\zeta_\parallel)}$, enters nothing at all at this order, by the gradient counting above, even though it does shape the transient fields.

1. Validity window

What we have built is a Green function of the polar-closed equations for $t > 0$, not a controlled approximation to the microscopic dynamics at the singular instant $t = 0$. Comparison with a particle simulation therefore holds only for times and Fourier modes satisfying

$$\begin{aligned} v_{\max}t + \sqrt{4D_{T,\max}t} &\ll d_{\rm wall}, \\ \frac{v^{(0)}k}{\mathscr{D}_{R,2}^{(0)}} \ll 1, \qquad \frac{2|\vec{p}|}{\rho} &\ll 1, \end{aligned} \tag{D4}$$

with $d_{\rm wall}$ the smallest distance from the initial support to the boundary, and the last condition evaluated where the fields are compared. Each inequality answers for one approximation: the first keeps the wall out, the second excludes scales on which the neglected nematic harmonic is not small, the third checks near-isotropy.

Inside this window the theory predicts the fields with no fitted transport coefficient, so any discrepancy that survives numerical refinement is a verdict on the closure itself. Outside it, the wall or the higher angular moments must be restored before a comparison means anything at all.

2. Reading parameters off the mode time constants (DDM)

Everything above is measurable, and the density half of it without tracking a single cell. Fourier-transforming a density field and watching each mode decay is exactly what differential dynamic microscopy does: it pulls the intermediate scattering function $f(k,t) = \langle e^{i\vec{k}\cdot(\vec{r}_t-\vec{r}_0)}\rangle$ out of the fluctuations of ordinary time-lapse images, averaged over $\sim 10^4$ cells [5, 6]. Within the polar closure that function *is* the density amplitude $A_k(t)$ of Eq. (C11).

This section runs the measurement backwards: from a decay curve to the microscopic combination behind it.

In a uniform landscape three decay channels are available at each k. The transverse polarity decays as a single exponential with

$$\tau_T(k) = \frac{1}{D_T^{(0)}k^2 + \mathscr{D}_R^{(0)}}, \tag{D5}$$

while for $k < k_c$ the coupled density and longitudinal polarity decay as a sum of two exponentials,

$$\tau_{\rm slow}(k) = \frac{1}{\mu_k - \sigma_k/2}, \qquad \tau_{\rm fast}(k) = \frac{1}{\mu_k + \sigma_k/2}. \tag{D6}$$

For $k > k_c$ the two merge into a single decay μ_k modulated at frequency $\omega_k = |\sigma_k|/2$, so k_c shows up directly as the wavenumber where relaxational modes give way to oscillatory ones.

Together these curves determine $(v^{(0)}, D_T^{(0)}, \mathscr{D}_R^{(0)})$, with one relation to spare. Their $k = 0$ intercept fixes $\mathscr{D}_R^{(0)}$; the slope of $1/\tau_T$ against k^2 fixes $D_T^{(0)}$; and $v^{(0)}$ follows from

$$\frac{1}{\tau_T(k)} - \frac{1}{\tau_{\rm fast}(k)} \;\longrightarrow\; \frac{(v^{(0)})^2}{2\mathscr{D}_R^{(0)}}\,k^2 \qquad (k \to 0), \tag{D7}$$

the slow channel providing the cross-check through $1/\tau_{\rm slow} \to D_{\rm eff}^{(0)}k^2$. These are precisely the quantities

a two-exponential fit to $f(k,t)$ returns, so the uniform-background part of the inversion asks for no new instrument.

One decomposition is out of reach here. Density and polarity cannot split $\mathscr{D}_R^{(0)} = D_R^{(0)} + \lambda^{(0)}(1-\mathscr{P}_{\rm turn})$ into its two pieces, because both enter it the same way. The nematic rate $\mathscr{D}_{R,2}^{(0)}$ can, since rotational diffusion damps harmonic n as n^2 while tumbling contributes $1-\mathscr{P}_{n,\rm turn}$. For the wrapped-normal kernel used here and $\mathscr{P}_{\rm turn}=0.3$, level-dependent rotational diffusion and tumbling drive the second harmonic in a ratio close to $3:1$.

Repeating the release in the landscape adds three scalars: the fast-rate shift,

$$\frac{1}{\tau_{\rm fast}} - \mathscr{D}_R^{(0)} \to \varepsilon\,\mathscr{D}_R^{(\eta)}\eta_1(\vec{r}), \tag{D8}$$

the spreading rate $4\langle D_{\rm eff}\rangle$, and the centroid velocity $\varepsilon\mathscr{V}\vec{\nabla}\eta_1$. Between them these return $(\mathscr{D}_R^{(\eta)}, \delta D_{\rm eff}, \kappa)$, with $\kappa = \mathscr{V} - \delta D'_{\rm eff}$.

Table III shows what is left over. Cases 2 and 4 share the relaxation shift, Cases 6–9 share the mobility, and Case 5 is missing altogether. Three numbers cannot carry nine rules; separating them needs either independent landscape components or an observable beyond density and polarity. The next two sections supply both.

Appendix E: Balance laws in detail

1. Solving for the six combinations

Six independent combinations exhaust what the nine rules can express. Four collect the contribution of one response channel each,

$$\begin{aligned} m_\chi &= \frac{v^{(0)}\big(\chi - \mathscr{D}_R^{(\zeta_\parallel)}\big)}{2\mathscr{D}_R^{(0)}}, \qquad m_v^{(\zeta_\parallel)} = \frac{v^{(\zeta_\parallel)}}{2}, \\ m_v^{(\eta)} &= \frac{v^{(0)}v^{(\eta)}}{2\mathscr{D}_R^{(0)}}, \qquad m_R^{(\eta)} = \frac{\big(v^{(0)}\big)^2\mathscr{D}_R^{(\eta)}}{2\big[\mathscr{D}_R^{(0)}\big]^2}, \end{aligned} \tag{E1}$$

together with the two jitter sensitivities $D_T^{(\eta)}$ and $D_T^{(\zeta_\parallel)}$. In this basis the transport coefficients are simple sums,

$$\begin{aligned} \mathscr{V} &= m_\chi + m_v^{(\zeta_\parallel)} + m_v^{(\eta)} - m_R^{(\eta)}, \\ \kappa &= m_\chi + m_v^{(\zeta_\parallel)} - m_v^{(\eta)} - D_T^{(\eta)}, \\ \mathscr{V}_v &= m_v^{(\zeta_\parallel)} + m_v^{(\eta)}, \end{aligned} \tag{E2}$$

with $\mathscr{V}_v$ the part of the drift carried by the two speed rules. Reading down Table IV solves for all six in order. The shift in the fast decay rate gives $m_R^{(\eta)}$. The polarity plateau read against the centroid gives $\mathscr{V}_v$, and hence $m_\chi = \mathscr{V} + m_R^{(\eta)} - \mathscr{V}_v$. The short-time exponents—or equivalently the tracked-cell lag shapes—split $m_v^{(\zeta_\parallel)}$ from $m_v^{(\eta)}$. The width gives $D_T^{(\eta)}$, and the part of it that does not scale with wavelength gives $D_T^{(\zeta_\parallel)}$.

TABLE IV. **A field guide: which measurement identifies which mechanism.** This is the paper's practical summary—each row is an experiment you can do, what it returns, and which rules it tells apart. The first block needs only time-lapse images of cell density, no tracking. The second needs the release watched until it stops changing. The third needs single-cell trajectories over a common lag. *Read together, the rows determine all six combinations of Eq.* (E1)*, and hence all nine sensitivities*—except two pairs (level–wander with level–tumble, aimed–wander with aimed–tumble) that only a measurement of nematic order can split.

Measurement	Returns	Rules it isolates
Density images alone		
Decay of $f(k,t)$, uniform background (DDM)	$v^{(0)}, D_T^{(0)}, \mathscr{D}_R^{(0)}$	—
Shift of the fast decay rate	level turning strength	2, 4
Centroid displacement, odd mode	$\mathscr{V}$	—
Pulse shape $T_\parallel$, even mode	$\mathscr{V}$ (no background)	—
Spread ΔS, even mode	κ	—
Spread, wavenumber-independent part	$\ell_{\rm run}v^{(\zeta_\parallel)} + 2D_T^{(\zeta_\parallel)}$	5, 7
Heading plateau, with the centroid	the speed rules' share of $\mathscr{V}$	3, 7
Short-time exponents of $\vec{P}, \vec{X}, S$	which overlap drives each	six groups
Population averages over the landscape		
Uptake rate $\dot{\mathscr{U}}$	κ	—
Steady accumulation $\rho_{\max}/\rho_{\min}$	$\kappa/D_{\rm eff}^{(0)}$	—
Divergence $\mathscr{K}$ from a null forager	squares of the above	detection cost
Tracked windows of common lag τ		
Projected velocity $V_\parallel(\tau)$	$\mathscr{V}_j$, and Φ_j	7 | 3,6,8,9 | 2,4
$\cos 2\alpha$ harmonic	$\chi + \mathscr{D}_R^{(\zeta_\parallel)}$	9 from 6, 8
$\delta\eta\cos\alpha$ harmonic	$v^{(\eta)}$, $\mathscr{D}_R^{(\eta)}$	3 from 2, 4
Information rate $\mathscr{I}(\tau)$	$(v^{(\eta)})^2, (v^{(\zeta_\parallel)})^2$	3, 7
Information rate as $\tau\to 0$	$(D_T^{(\eta)})^2, (D_T^{(\zeta_\parallel)})^2$	1, 5

The four numbers of Eq. (11) close on one more, the outward polarity $\Omega = \int \vec{x}\cdot\vec{p}\,{\rm d}^2x$—how strongly cells point away from the release point, and the only channel by which swimming feeds spreading. Every flux in Eqs. (10) is a divergence, so inside the wall-free window it integrates away, leaving

$$\dot{\vec{P}} = -\mathscr{D}_R^{(0)}\vec{P} + \frac{\chi - \mathscr{D}_R^{(\zeta_\parallel)}}{2}\vec{\mathscr{G}}_\rho - \mathscr{D}_R^{(\eta)}\vec{\mathscr{E}}_p, \tag{E3a}$$

$$\begin{aligned} \dot{\Omega} = &-\mathscr{D}_R^{(0)}\Omega + Nv^{(0)} + v^{(\eta)}\mathscr{E}_\rho + v^{(\zeta_\parallel)}\mathscr{G}_p \\ &- \mathscr{D}_R^{(\eta)}\mathscr{E}_p + \frac{\chi - \mathscr{D}_R^{(\zeta_\parallel)}}{2}\mathscr{E}_\rho, \end{aligned} \tag{E3b}$$

$$N\dot{\vec{X}} = v^{(0)}\vec{P} + v^{(\eta)}\vec{\mathscr{E}}_p + \frac{v^{(\zeta_\parallel)}}{2}\vec{\mathscr{G}}_\rho, \tag{E3c}$$

$$N\dot{S} = 4ND_T^{(0)} + 2v^{(0)}\Omega$$

$$+ 2v^{(\eta)}\mathscr{C}_p + v^{(\zeta_\parallel)}\mathscr{C}_\rho + 4D_T^{(\eta)}\mathscr{E}_\rho + 4D_T^{(\zeta_\parallel)}\mathscr{G}_p. \tag{E3d}$$

The six overlaps are named for the field they weight (ρ or $\vec{p}$) and what they read—the level $\mathscr{E}$, the gradient $\mathscr{G}$, or the gradient moment about the release point $\mathscr{C}$:

$$\begin{aligned} \mathscr{E}_\rho &= \int \rho\,\delta\eta, & \vec{\mathscr{G}}_\rho &= \int \rho\,\vec{\nabla}\eta, \\ \vec{\mathscr{E}}_p &= \int \delta\eta\,\vec{p}, & \mathscr{G}_p &= \int \vec{p}\cdot\vec{\nabla}\eta, \\ \mathscr{C}_\rho &= \int \rho\,\vec{x}\cdot\vec{\nabla}\eta, & \mathscr{C}_p &= \int \delta\eta\,(\vec{x}\cdot\vec{p}). \end{aligned} \tag{E4}$$

Those taken against the density start finite; those taken against the polarity start at zero and grow like t, which is the origin of the hierarchy in Table II.

2. Uniform background

Set $\varepsilon = 0$ and every overlap vanishes. Then $\vec{P}$ and $\vec{X}$ stay zero by symmetry, and the remaining pair integrates in closed form,

$$\begin{aligned} \Omega^{(0)} &= \frac{Nv^{(0)}}{\mathscr{D}_R^{(0)}}\left(1 - e^{-\mathscr{D}_R^{(0)}t}\right), \\ S^{(0)}(t) - S(0) &= 4D_T^{(0)}t \\ &\quad + \frac{2\big(v^{(0)}\big)^2}{\big[\mathscr{D}_R^{(0)}\big]^2}\left(\mathscr{D}_R^{(0)}t - 1 + e^{-\mathscr{D}_R^{(0)}t}\right). \end{aligned} \tag{E5}$$

Outward alignment saturates at $Nv^{(0)}/\mathscr{D}_R^{(0)}$ within one orientational time, and at the same moment the spread crosses over from $4D_T^{(0)}t + \big(v^{(0)}t\big)^2$ to $4D_{\text{eff}}^{(0)}t - 2\big(v^{(0)}\big)^2/[\mathscr{D}_R^{(0)}]^2$: ballistic, then diffusive. One spreading curve therefore fixes $D_T^{(0)}$, $v^{(0)}$ and $\mathscr{D}_R^{(0)}$ before any landscape is applied—a useful calibration to have in hand.

3. Saturation, refinements and the polarity plateau

Take a single landscape mode of wavevector $\vec{q}$. Its overlaps collapse straight onto the Green amplitudes of Eq. (C11). For an odd mode $\varepsilon\sin(\vec{q}\cdot\vec{x})$ about the point release of Sec. C 3, $\vec{\mathscr{G}}_\rho = N\hat{q}\,qA_q(t)$ and $\vec{\mathscr{E}}_p = N\hat{q}\,B_q(t)$ with $\mathscr{E}_\rho = \mathscr{C}_\rho = \mathscr{C}_p = \mathscr{G}_p = 0$; an even mode does the reverse; and a release of finite width multiplies both by its own transform.

Now A_q and B_q are combinations of $e^{-t/\tau_{\text{slow}}(q)}$ and $e^{-t/\tau_{\text{fast}}(q)}$, and

$$\int_0^t e^{-\mathscr{D}_R^{(0)}(t-t')}e^{-st'}\,\mathrm{d}t' = \frac{e^{-st} - e^{-\mathscr{D}_R^{(0)}t}}{\mathscr{D}_R^{(0)} - s}, \tag{E6}$$

so every observable is a fixed combination of three exponentials, at rates already tabulated in Sec. D 2. The macroscopic observables bring no time scale of their own.

Two of those rates matter. Alignment equilibrates at $\mathscr{D}_R^{(0)}$. The mode itself then decays at $1/\tau_q = D_{\text{eff}}^{(0)}q^2$, because once the pulse has spread across a wavelength the gradient it feels averages away.

This has a practical consequence. A release in a structured landscape reports a *displacement*, not a velocity: there is no slope to fit. What it reports is the integral of Eqs. (E3) over the whole life of the mode, Eq. (14) with $\Delta S = S - S^{(0)}$. This is the section's central result: *the centroid saturates on the drift, the spread saturates on the mobility.*

Section VI argued that $\mathscr{V}$ and κ are different coefficients answering different questions. Here one release returns both, from two landscape components of opposite parity, and their difference is $\delta D'_{\text{eff}}$. The second line also says something one might not have guessed: a population that accumulates up-gradient ($\kappa > 0$) ends up *narrower* than it would in a uniform landscape, by exactly four times the accumulation length $\kappa/D_{\text{eff}}^{(0)}q^2$.

Neither expression needs the adiabatic elimination that produced Eq. (D3). Integrating $\dot{\vec{P}} = -\mathscr{D}_R^{(0)}\vec{P} + (\text{source})$ from zero to infinity gives $\int\vec{P}\,\mathrm{d}t = \int(\text{source})\mathrm{d}t/\mathscr{D}_R^{(0)}$ whatever the transient does, and likewise for Ω; Eq. (14) is exact at $\mathcal{O}(\varepsilon)$.

The same integral supplies two refinements, and each carries information rather than merely correcting a number. First, a release of width σ_0 in a landscape of finite wavenumber replaces $1/D_{\text{eff}}^{(0)}$ by $e^{-q^2\sigma_0^2/2}/D_{\text{eff}}(q)$ with

$$D_{\text{eff}}(q) = D_T^{(0)} + \frac{\big(v^{(0)}\big)^2}{2\big(D_T^{(0)}q^2 + \mathscr{D}_R^{(0)}\big)}, \tag{E7}$$

the zero-frequency transfer of the propagator itself. Second, the spread carries a term that the gradient expansion behind Eq. (D2) throws away,

$$\Delta S_\parallel(\infty) = -\frac{\varepsilon\,\ell_{\text{run}}}{D_{\text{eff}}^{(0)}}\left[\ell_{\text{run}}\,v^{(\zeta_\parallel)} + 2D_T^{(\zeta_\parallel)}\right], \quad \ell_{\text{run}} = \frac{v^{(0)}}{\mathscr{D}_R^{(0)}}, \tag{E8}$$

smaller than the mobility term by $(q\ell_{\text{run}})^2$ and carried *only* by the two directional flux responses. It is the one long-time observable Case 5 reaches at all. The two terms are cleanly separable because the mobility term scales as q^{-2} and this one does not: repeat the release at two wavelengths and read off both.

The net polarity does not saturate. It rises over $1/\mathscr{D}_R^{(0)}$ to a quasi-static plateau and then decays with the mode,

$$\frac{\vec{P}}{N} \simeq \frac{\varepsilon q}{v^{(0)}}\big[\mathscr{V} - \mathscr{V}_v\big]e^{-t/\tau_q}\,\hat{q}, \qquad \mathscr{V}_v = \frac{v^{(\zeta_\parallel)}}{2} + \frac{v^{(0)}v^{(\eta)}}{2\mathscr{D}_R^{(0)}}, \tag{E9}$$

where $\mathscr{V}_v$ is the part of the drift carried by the two speed responses. Alignment thus reports the drift *minus* the speed channel, which makes sense: a rule that moves density merely by making cells faster on one side never creates net alignment.

Reading the polarity plateau against the centroid displacement therefore isolates $\mathscr{V}_v$, and Table II already splits its two members by their exponents. Three curves, four coefficients: $\mathscr{V}$, κ, $\mathscr{V}_v$, and the directional pair of Eq. (E8).

4. Balance laws for the shape

The traceless part $\mathbf{T}$ of the second moment, Eq. (11), is the elongation of the released cloud, and it costs nothing extra to measure. Repeating the integration by parts with this traceless weight gives one more pair of balance laws,

$$N\dot{T}_{ij} = v^{(0)}\hat{\Omega}_{ij} + v^{(\eta)}\hat{\mathscr{C}}_{p,ij} + \tfrac{1}{2}v^{(\zeta_\parallel)}\hat{\mathscr{C}}_{\rho,ij}, \tag{E10a}$$

$$\dot{\hat{\Omega}}_{ij} = -\mathscr{D}_R^{(0)}\hat{\Omega}_{ij} + \tfrac{1}{2}v^{(\zeta_\parallel)}\hat{\mathscr{G}}_{p,ij} - \mathscr{D}_R^{(\eta)}\hat{\mathscr{C}}_{p,ij} + \tfrac{1}{2}\big(\chi - \mathscr{D}_R^{(\zeta_\parallel)}\big)\hat{\mathscr{C}}_{\rho,ij}, \tag{E10b}$$

in which $\hat{\Omega}_{ij}$ is the traceless outward-polarity moment and $\hat{\mathscr{C}}_\rho$, $\hat{\mathscr{C}}_p$, $\hat{\mathscr{G}}_p$ are the traceless companions of $\mathscr{C}_\rho$, $\mathscr{C}_p$ and $\mathscr{G}_p$, obtained by replacing $x_i\partial_j\eta$ with $x_i\partial_j\eta + x_j\partial_i\eta - \delta_{ij}\vec{x}\cdot\vec{\nabla}\eta$, and so on.

Two things follow at once. *No translational-diffusivity response appears.* A diffusive flux is a gradient, and a gradient feeds only the trace, so Cases 1 and 5 cannot shape a pulse at all. And the uniform background is exactly isotropic, so $\mathbf{T} \equiv \vec{0}$ at $\varepsilon = 0$: unlike S, the shape has no background to subtract, and the whole measured signal is the response.

What drives the shape is the anisotropy of the landscape's curvature: for a compact release, $\hat{\mathscr{C}}_{\rho,ij} \simeq \frac{1}{2}\langle|\vec{x}|^2\rangle(\partial_i\partial_j\eta - \frac{1}{2}\delta_{ij}\nabla^2\eta)$ evaluated at the release point. A landscape whose Hessian happens to be isotropic there gives no shape signal whatever its gradient. So this observable asks for a release where the curvature has a direction—which is exactly what a single plane-wave mode provides.

On the even mode of Sec. VII the shape saturates on *the drift*, Eq. (15) with $T_\parallel$ the component along $\hat{q}$—exactly half of Eq. (14) with $-\kappa$ replaced by $\mathscr{V}$.

So one even landscape mode returns both coefficients at once, from the two parts of a single tensor: the trace gives κ, the traceless part gives $\mathscr{V}$, and no mode of the opposite parity is needed. Where the two disagree, the population is being pushed by a spatially varying diffusivity rather than steered.

Appendix F: Foraging and information: details

1. One landscape mode

On a single mode, $\eta = \eta^{(0)} + \varepsilon\cos(\vec{q}\cdot\vec{x} + \varphi)$, the harvest law closes. There $\langle\nabla^2\eta\rangle = -q^2(\mathcal{U} - \eta^{(0)})$ exactly, and $\langle|\vec{\nabla}\eta|^2\rangle \to \varepsilon^2 q^2/2$ once the pulse has spread across a wavelength, so $\mathcal{U}$ relaxes as a single exponential from whatever the release point offered to

$$\mathcal{U}(\infty) - \eta^{(0)} = \frac{\kappa\varepsilon^2}{2D_{\text{eff}}^{(0)}}, \qquad \tau_\ell = \frac{1}{D_{\text{eff}}^{(0)}q^2} = \frac{\ell^2}{4\pi^2 D_{\text{eff}}^{(0)}}. \tag{F1}$$

The two halves come apart usefully. The gain does not depend on q at all: a fine landscape and a coarse one of the same contrast are worth the same. The time needed to collect it does depend on q, as ℓ^2. A fine-grained landscape is not richer, only quicker.

2. Telling two strategies apart

Suppose two mechanisms—or one mechanism and a null forager—produce densities ρ_i and ρ_j. How hard are they to tell apart? The natural measure is the Kullback–Leibler divergence per cell,

$$\mathscr{K}_{ij} = \frac{1}{N}\int \rho_i \ln\frac{\rho_i}{\rho_j}\,\mathrm{d}^2x = \frac{1}{2N}\int \frac{(\rho_i - \rho_j)^2}{\rho^{(0)}}\,\mathrm{d}^2x + \mathcal{O}(\varepsilon^3), \tag{F2}$$

the linear term vanishing because both distributions carry the same mass. Everything now follows from one fact: $\mathscr{K}$ is quadratic where the fields are linear.

Every exponent therefore doubles. A rule whose density response grows as t^n in Eq. (C21) becomes distinguishable as t^{2n}. And M independently tracked cells accumulate a log-likelihood ratio $M\mathscr{K}$, so a detection needs

$$M \gtrsim \mathscr{K}^{-1} \sim t^{-2n}. \tag{F3}$$

That is the sampling floor of Sec. X in closed form, and it settles the experimental trade. At fixed cost Mt the evidence grows as t^{2n-1}, so watching longer always beats tracking more. For Cases 2 and 4, where $n = 3$, doubling the window is worth 32 times the cells.

The same expression states the degeneracies exactly. Cases 2 and 4 generate identical fields, so $\mathscr{K}_{24} = 0$ at every time and every ensemble size. Cases 6, 8 and 9 generate collinear fields, differing only by the scalar $\chi - \mathscr{D}_R^{(\zeta_\parallel)}$, so no release can separate the mechanism from its amplitude. Neither statement is about statistical power; both are about the model.

3. The backward generator

Both follow from the backward generator adjoint to Eq. (A8),

$$\mathscr{L}f = v\,\hat{n}\cdot\vec{\nabla}f + D_T\nabla^2 f + \chi\zeta_\perp\partial_\theta f + D_R\partial_\theta^2 f + \lambda\int_0^{2\pi}\widehat{\mathscr{T}}(\Delta\theta)\big[f(\vec{r},\theta+\Delta\theta) - f(\vec{r},\theta)\big]\mathrm{d}(\Delta\theta), \tag{F4}$$

in which a state-dependent Itô diffusivity multiplies $\nabla^2 f$, where in the forward equation it sat inside $\nabla^2(D_T P)$. That difference is not bookkeeping. Population redistribution and single-trajectory drift are different statements, and the two translational-diffusivity rules are about to prise them apart.

4. Finite-lag harmonics

Resolving the window by its own initial heading buys more still. Writing $\delta\eta$ for the level the cell was released

at, the conditional profile is

$$V_{\parallel}(\delta\eta,\alpha;\tau) = v^{(0)}\cos\alpha\, W(u) + \mathscr{V}_j\, g\, \Phi_j + b_2(\tau)\cos 2\alpha + b_\eta(\tau)\,\delta\eta\cos\alpha, \tag{F5}$$

with $u = \mathscr{D}_R^{(0)}\tau$, the second term the drift found above, and

$$b_{2;6,8,9} = -\frac{v^{(0)}g}{2}\big(\chi + \mathscr{D}_R^{(\zeta_\parallel)}\big)\Theta(\tau),$$
$$b_{2;7} = \frac{v^{(\zeta_\parallel)}g}{2}\, W\Big(\mathscr{D}_{R,2}^{(0)}\tau\Big), \tag{F6}$$
$$b_{\eta;3} = v^{(\eta)}W(u), \qquad b_{\eta;2,4} = -\frac{v^{(0)}\mathscr{D}_R^{(\eta)}}{\mathscr{D}_R^{(0)}}\Lambda(u),$$

where $\Lambda(u) = [1-(1+u)e^{-u}]/u$ and $\Theta = [W(\mathscr{D}_{R,2}^{(0)}\tau) - W(\mathscr{D}_R^{(0)}\tau)]/(\mathscr{D}_R^{(0)} - \mathscr{D}_{R,2}^{(0)})$, both $\simeq \tau/2$ at short lag.

Two degeneracies break here. The drift carries $\chi - \mathscr{D}_R^{(\zeta_\parallel)}$; the *second* harmonic carries $\chi + \mathscr{D}_R^{(\zeta_\parallel)}$, and it relaxes at the nematic rate $\mathscr{D}_{R,2}^{(0)}$. Two independent combinations separate steering from heading-dependent turning—the one degeneracy Table II cannot break. It is no accident that the missing information sits in exactly the harmonic the polar closure threw away.

Likewise b_η separates a level speed response from a level turning response, by lag shape as well as by sign. As $\tau \to 0$, $b_{\eta;3} \to v^{(\eta)}$, returning the bare sensitivity with no transport coefficient attached, while $b_{\eta;2,4}$ vanishes with the lag.

5. Nutrient gain closes the loop

Project on $\vec{\nabla}\eta$ instead of $\hat{g}$ and the same window returns the nutrient gain $\mathbb{E}[\eta(\vec{r}_{t+\tau}) - \eta(\vec{r}_t)]/\tau$. Averaged over the whole population this is nothing but a difference quotient of the uptake of Sec. VIII. Expanding it,

$$\dot{\mathscr{U}} = \mathscr{V}\big\langle|\vec{\nabla}\eta|^2\big\rangle + \big\langle D_{\rm eff}\nabla^2\eta\big\rangle, \tag{F7}$$

the first term being the single-trajectory drift, the second the spreading. On a bounded landscape $\langle\eta_1\nabla^2\eta_1\rangle = -\langle|\vec{\nabla}\eta_1|^2\rangle$, which turns $\mathscr{V}$ into κ and recovers Eq. (17) exactly. The individual drift is $\mathscr{V}$, the harvest is κ, and the landscape's own curvature makes up the difference.

6. Finite-lag information

At finite lag the same two channels persist and the noise itself crosses over. The projected spread is $\langle(Y^\tau)^2\rangle = 2D_\parallel(\tau)\tau$ with

$$D_\parallel(\tau) = D_T^{(0)} + \frac{\big(v^{(0)}\big)^2}{2\mathscr{D}_R^{(0)}}\,\Phi\Big(\mathscr{D}_R^{(0)}\tau\Big), \tag{F8}$$

the same Φ again, now carrying the noise from the bare $D_T^{(0)}$ at short lag to $D_{\rm eff}^{(0)}$ at long lag. Writing $\bar\sigma^2$ for the state-averaged conditional variance, the response is

$$\Delta\mathscr{J}(\tau) \simeq \frac{\tau\big[b_2^2 + b_\eta^2\,\mathrm{Var}(\delta\eta)\big]}{4\bar\sigma^2} + \frac{\mathrm{Var}\big[\Delta\sigma^2\big]}{4\tau\bar\sigma^4}, \tag{F9}$$

whose two terms are the mean and variance channels, and which reproduces Eq. (F10) as $\tau \to 0$.

One last observation ties this section to the previous one. Every entry in Eq. (F10) is *quadratic* in the response, because information is a quadratic functional of the fields exactly as the divergence $\mathscr{K}$ of Sec. VIII is. The sample-size law of Eq. (F3) therefore applies unchanged: the number of tracked windows a detection needs falls as the square of the sensitivity, and doubling the lag beats doubling the ensemble wherever the response is still growing.

Evaluated on the isotropic release, the four legible responses give

$$\Delta\mathscr{J}_3 = \frac{(v^{(\eta)})^2\,\mathrm{Var}(\delta\eta)}{8D_T^{(0)}},\ \Delta\mathscr{J}_7 = \frac{(v^{(\zeta_\parallel)})^2\big(3\langle g^2\rangle - 2\langle g\rangle^2\big)}{32D_T^{(0)}},$$
$$\Delta\mathscr{J}_1 = \frac{(D_T^{(\eta)})^2\,\mathrm{Var}(\delta\eta)}{4\big[D_T^{(0)}\big]^2\tau},\ \Delta\mathscr{J}_5 = \frac{(D_T^{(\zeta_\parallel)})^2\langle g^2\rangle}{8\big[D_T^{(0)}\big]^2\tau}, \tag{F10}$$

Appendix G: Numerical methods

1. Choosing the landscape

The landscape is a single cosine, $\eta = 1 + \varepsilon\cos(\vec{q}\cdot\vec{x} + \varphi)$ (Fig. 6), so that every prediction of Secs. IV and VIII applies verbatim rather than through a spectral sum.

Three requirements pin it down, and they conflict. Only the zeros of J_1 make the disk average of the modulation vanish identically, so the admissible wavenumbers form a ladder. Climbing it, the release window $e^{-q^2\sigma_0^2/2}$ falls, the closure error $\big(v^{(0)}q/\mathscr{D}_{R,2}^{(0)}\big)^2$ rises, and the mode lifetime τ_q shrinks toward the run time. The second rung, $q\mathscr{R} = 7.0156$, is the only one where all three are comfortable at once: 0.78, 3.8%, and $\tau_q = 0.0102$—twice the run. The phase $\varphi = \pi/4$ then puts equal odd and even content about the release, exciting centroid and spread together, and $\varepsilon = 0.25$ keeps η positive everywhere.

2. Calibrating the sensitivities

The nine sensitivities are calibrated rather than inherited from any organism, since what this paper compares is rules. The choice is still constrained. Each response is linear in its own coefficient, so the only freedom is where to place a case along its ray (u_ρ, u_p) of density and polarity response per unit sensitivity, and the ratio u_ρ/u_p is fixed by the physics. It spans a factor 19 across the nine, so matching the density panels exactly would spread the polarity panels by 19, and the reverse. Setting each sensitivity to $g/\sqrt{u_\rho u_p}$ splits the difference, leaving both

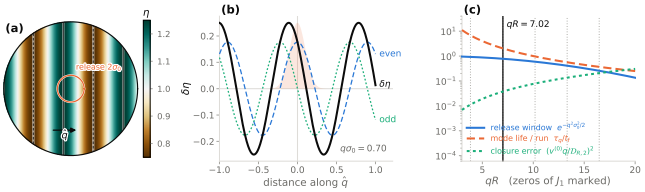


FIG. 6. Why this mode. (a) The landscape on the arena and (b) its cut along $\hat{q}$, both reproduced from Fig. 2 for reference. (c) Admissible modes are the zeros of J_1 (dotted), where the disk average of $\delta\eta$ vanishes. Climbing that ladder, the release window falls, the closure error rises and the mode lifetime shrinks toward the run time; the second zero (solid line) is the only rung where all three are comfortable.

fields spread by only $\sqrt{19} = 4.3$—close enough for one linear colour scale to carry both. Cases 4 and 8 are then held at $\lambda^{(\cdot)} = D_R^{(\cdot)}/(1 - \mathscr{P}_{\rm turn})$ and Case 9 at $\chi = D_R^{(\zeta_\parallel)}$, which turns the exact degeneracies $2 \equiv 4$ and $6 \equiv 8$ into numerical checks.

Visibility costs linearity. Cases 2 and 4 need the largest fractional modulation of the coefficient they act on, $D_R^{(\eta)}\varepsilon/D_R^{(0)} = 0.24$, against 0.12 for Case 5 and below 0.09 for the rest, so their own $\mathcal{O}(\varepsilon^2)$ error is the largest of the nine. The quantitative agreement quoted below therefore rests on the strong rules; the weak ones are read as shapes.

3. Scope

Two further checks address the analytic structure rather than any single case. Integrating the closed system directly in a linear landscape, with no adiabatic elimination anywhere, recovers $D_{\rm eff}^{(0)}$ and the centroid velocity of Eq. (13) for all nine rules—Cases 1 and 5 giving an identically vanishing drift, as predicted. The divergence $\mathscr{K}$ between two strategies is quadratic where the fields are linear, so every transient exponent doubles and a detection needs $M \gtrsim \mathscr{K}^{-1}$ tracked cells; measured exponents are 1.95–1.97, 3.91–3.96 and 5.94 against the predicted 2, 4 and 6. At fixed cost Mt the evidence grows as t^{2n-1}, so watching longer always beats tracking more.

The scope of all this is three parameter sets and three landscapes, each inside its own wall-free window, which must be rechecked for any other.

Appendix H: Remaining response mechanisms

Figure 7 compares Cases 2–9 using the parameters and initial condition of Fig. 2. Each row is one mechanism; the columns are particles, the polar PDE and the Fourier solution; every panel is at the common time $t = 5 \times 10^{-3}$ ($t\mathscr{D}_R^{(0)} = 1.78$). One linear scale is shared by all density panels and another by all polarity panels, with the same limits and spatial window as Fig. 2 and both supplementary videos.

Reading the rows against that common scale is the point of the figure. With the sensitivities calibrated as in Sec. X, no mechanism is off-scale and none is invisible, so what differs between rows is the *shape* of the response, not its size. Cases 2 and 4 remain indistinguishable, as do Cases 6, 8 and 9 in density—the model's own degeneracies, reproduced by all three solvers. Supplementary Video 2 shows the full evolution.

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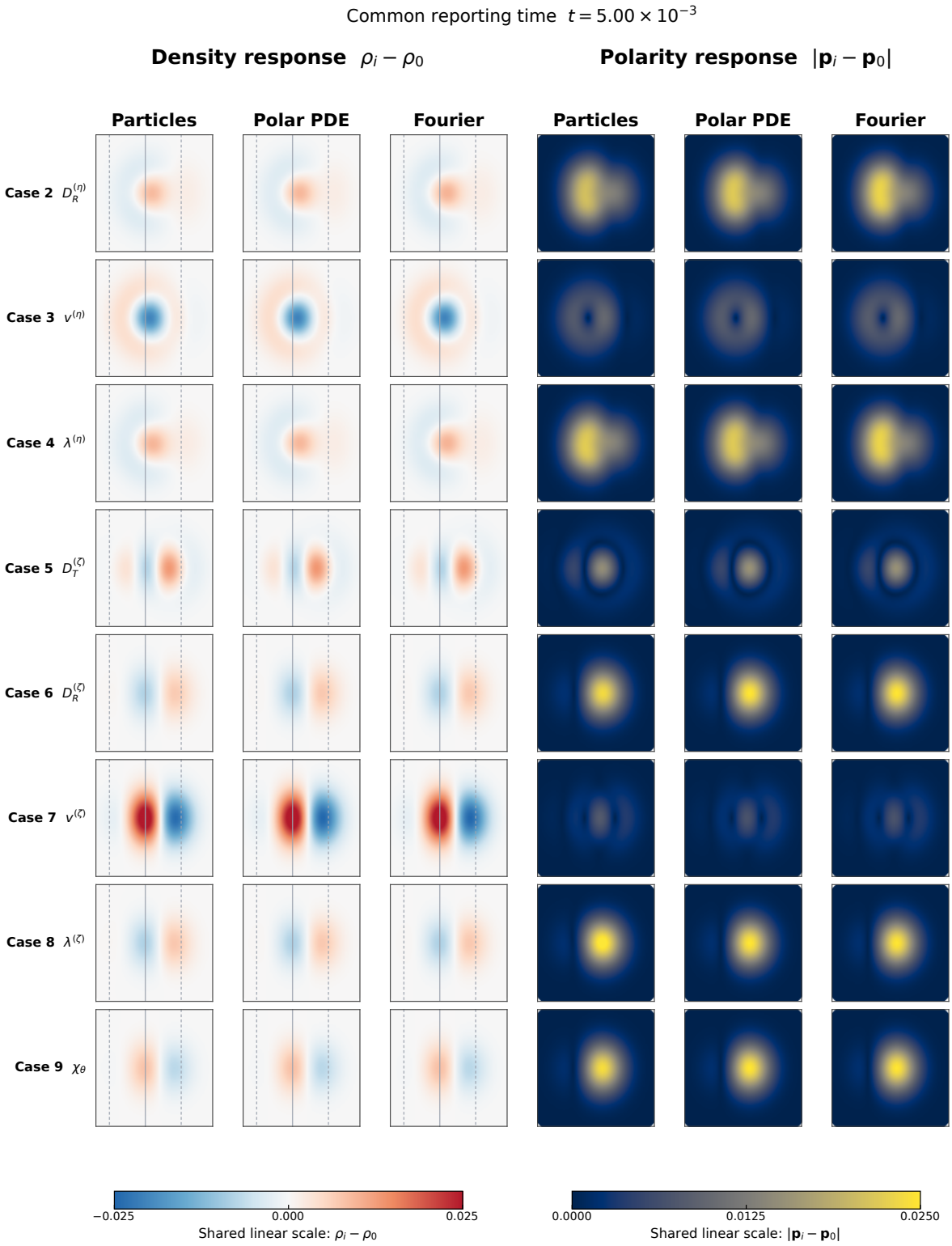


FIG. 7. Three descriptions reproduce the eight remaining response mechanisms, all at $t = 5 \times 10^{-3}$. Columns are particles, the polar PDE and the Fourier solution, for density (left) and polarity (right). Each observable has one shared linear scale, so what differs between rows is the shape of the response, not its size. Lines mark nutrient crests and troughs as in Fig. 6.